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Degree Thesis in Microdata Analysis
Level: Master of Science (MSc) in Business Intelligence
Congestion and Traffic Flow Analysis Using
Trafikverket Sensor Data

Author: Md Nazmul Islam Razib(h24mdnra@du.se), Md Ariful Ahsan(h24mdaah@du.se)

School: School of Information and Engineering

Supervisor: Johan Håkansson (jhk@du.se), Vijay Paidi (vpp@du.se)

Examiner: Mia Xiaoyun Zhao (xzh@du.se)

Subject/main field of study: Microdata Analysis (Business Intelligence)

Course code: MI4002

Credits: 15

Date of examination: 14.01.2026

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Abstract:

The increasing digitalization of road infrastructure has led to the continuous generation of large-scale traffic sensor data. Despite this abundance of information, the strategic use of historical traffic datasets remains limited due to challenges related to data extraction, structural inconsistency, and analytical integration. In particular, publicly available traffic data from the Swedish Transport Administration (Trafikverket) are distributed in semi-structured formats that require preprocessing before meaningful analysis can be conducted.

This thesis develops a reproducible end-to-end analytical framework that transforms traffic data into structured, analysis-ready datasets for congestion and environmental assessment. A Python-based web crawling system automates large-scale data acquisition, while the processed data are stored in a centralized PostgreSQL database to ensure consistency, scalability and reproducibility.

This study utilizes a scalable analytical framework that adapts to datasets of different sizes, focusing on methodological rigor rather than the quantity of data. The analysis is based on 5,137 observations from eight measurement points across four Swedish counties between 1998 and 2023. The study constructs and validates three congestion indicators: Speed Ratio (SR), Delay Index (DI), and Speed Variability. The analysis identifies persistent moderate congestion, systematic differences between passenger cars and heavy vehicles, and pronounced peak-hour dynamics.

To assess environmental impacts, congestion metrics are integrated with speed-dependent CO₂ emission modeling based on the COPERT methodology. The results indicate a strong non-linear relationship between congestion severity and emission intensity, with severe congestion producing disproportionately higher per-vehicle emissions. Spatial clustering of congestion-emission hotspots further highlights priority areas for targeted interventions.

To improve data accessibility, the framework integrates a Model Context Protocol (MCP) server prototype that enables users to query and explore the structured historical traffic datasets using natural language.

Overall, the proposed framework integrates data engineering, congestion modeling, environmental assessment, and AI-assisted querying into a scalable workflow that supports evidence-based transportation planning and sustainable mobility policy.

Keywords: *Intelligent Transportation Systems (ITS), Traffic Congestion Analysis, Speed Ratio, Delay Index, CO₂ Emissions Modeling, Flow-Speed Regimes, Trafikverket, Data Engineering Pipeline, Web Scraping, PostgreSQL, Model Context Protocol (MCP), AI-Assisted Analytics, Sustainable Transportation Planning.*

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1. Introduction

The rapid expansion of Intelligent Transportation Systems (ITS) has transformed traffic monitoring and management, but the analytical use of historical sensor data remains limited. Across national road networks, roadside sensors and automated counting stations continuously record vehicle speeds, traffic volumes, vehicle classifications, and temporal patterns. These systems generate vast amounts of high-resolution data, creating new opportunities for systematic and data-driven traffic analysis (Michele, 2024). However, most existing applications of traffic data focus primarily on short-term operational objectives. Navigation platforms and congestion monitoring tools rely primarily on real-time data streams to support route optimization and traveler decision-making. In contrast, comparatively little attention has been devoted to reproducible historical analyses based on publicly accessible sensor datasets. In particular, the integration of reproducible historical data extraction, congestion analysis by vehicle type, and CO₂ emission assessment remains insufficiently explored.

Simultaneously, cities and national authorities are increasingly prioritizing smart mobility and sustainable transport policies (European Union, 2021). In this context, historical traffic data has become particularly valuable, enabling researchers and policymakers to evaluate long-term congestion trends, assess infrastructure performance, and examine the environmental consequences of evolving traffic dynamics. Despite its substantial potential, practical challenges continue to limit the effective utilization of historical datasets.

In Sweden, the Swedish Transport Administration (Trafikverket) operates an extensive monitoring network across major highways and regional roads (Trafikverket, 2026). Traffic data are publicly accessible through the Trafikverket website, while selected real-time traffic information is also available via the public API (The Swedish Transport Administration's open API for traffic information, 2026). However, no dedicated public API exists for the systematic extraction of historical traffic data. Historical records are accessible only through the website interface (Trafikverket Road traffic flow map, 2026), where data are presented in graphical or tabular formats distributed across multiple URLs and lacking standardized schemas. Consequently, data collection requires substantial manual

effort, is time-consuming, and increases the risk of duplication and inconsistencies. Furthermore, the semi-structured and fragmented nature of the data, combined with limited machine readability, reduces their usability for advanced analytical studies.

To address these limitations and facilitate large-scale historical analysis, this thesis proposes a structured data pipeline designed to extract historical traffic data from Trafikverket with minimal human intervention. The proposed extraction framework is designed to automate data extraction, reduce manual effort, and minimize duplicate records. By transforming semi-structured web-based records into standardized analytical datasets, the pipeline enhances reproducibility and significantly lowers the barrier to historical traffic research.

Using the extracted dataset, the study conducts an empirical analysis based on 5,137 observations collected from eight measurement points across four Swedish counties between 1998 and 2023. A structured analytical framework is developed to evaluate traffic congestion and its environmental impacts. Three congestion indicators are constructed and analyzed: Speed Ratio (SR), Delay Index (DI), and Speed Variability. Free-flow speed is estimated using the 95th percentile methodology, providing a robust and consistent benchmark for congestion assessment. The Delay Index (DI) is defined as a normalized speed loss metric representing the proportional reduction in observed speed relative to free-flow conditions. This formulation differs from traditional travel-time-ratio-based definitions of delay; it is derived solely from speed observations and is therefore referred to here as a speed-based DI. It provides an intuitive measure of congestion by quantifying the deviation of observed speeds from optimal, uncongested conditions. A higher DI value indicates greater delay and reduced traffic efficiency, while values closer to zero represent near free-flow operation. Negative values may occur when observed speeds exceed the percentile-based free-flow benchmark.

In transportation research, such normalized measures are commonly used to enable comparisons across different road types, traffic conditions, and vehicle categories, as they remove the influence of absolute speed differences. The Delay Index is particularly useful for identifying congestion severity and evaluating traffic performance, as it captures the relative impact of congestion rather than relying

solely on raw speed measurements. Consequently, it serves as an effective metric for both spatial and temporal analyses of traffic conditions. Mathematically, the Delay Index can be expressed as the relative difference between free-flow speed and observed speed, normalized by free-flow speed, allowing consistent interpretation across varying traffic environments.

Beyond mobility impacts, the analysis integrates congestion indicators with speed-dependent CO₂ emission modeling based on the COPERT methodology (Ntziachristos, Gkatzoflias, Kouridis, & Samaras, 2009). The results suggest a non-linear relationship between congestion severity and emission intensity. Severe congestion generates disproportionately higher per-vehicle emissions, highlighting the substantial environmental costs associated with traffic breakdown conditions. Spatial hotspot analysis further identifies congestion-emission hotspots, suggesting that targeted interventions at critical bottlenecks may yield environmental benefits. (Zhang, Batterman, & Dion, 2011). Building upon these indicators, the study investigates spatiotemporal congestion patterns, identifying persistent hotspots, peak-hour dynamics, and systematic differences between passenger cars and heavy vehicles. The findings indicate that heavy vehicles consistently experience lower Speed Ratios and contribute disproportionately to congestion severity. Flow-speed regime analysis reveals clear transitions between uncongested, transitional, and breakdown states, providing deeper insight into capacity utilization and traffic instability.

In parallel with developments in transportation analytics, advances in artificial intelligence have expanded the capabilities of large language models (LLMs). Nevertheless, these models remain constrained by static training data and limited access to structured external information (Mousavi & Riccardi, 2025). The Model Context Protocol (MCP) addresses these limitations by providing a standardized and secure interface that enables LLMs to retrieve real-time context and perform structured operations. MCP follows a client–server architecture comprising three primary components: (i) the MCP host (e.g., an AI application interface), (ii) the MCP client embedded within the host to manage connections, and (iii) the MCP

server, which exposes tools, datasets, and workflows through schema-defined interfaces (Model Context Protocol Working Group, 2024).

Although MCP has recently gained attention in domains such as real-time analytics and tool-augmented AI systems, its application in traffic data analysis remains largely unexplored. A recently introduced MCP server, developed by TomTom (TomTom MCP Server, 2026), provides real-time traffic insights. However, systematic integration of historical traffic datasets into MCP-based analytical workflows has received limited scholarly attention.

To support reproducibility and illustrate the potential of AI-assisted traffic analysis, this thesis includes a demonstration MCP server that exposes structured historical traffic datasets as standardized resources. The server enables basic querying of vehicle counts, speed distributions by vehicle type, and spatiotemporal traffic patterns based on user-defined parameters such as location and date. While the MCP client integration (e.g., Claude Desktop) shows how natural language queries can be applied, this component is presented as a practical prototype rather than a primary evaluated contribution. Its role is to complement the core research on congestion modeling, emission analysis, and data structuring by illustrating how standardized access mechanisms can facilitate future AI-enabled decision-support systems.

Overall, this work makes two primary contributions: a reproducible historical traffic data pipeline and an empirical analysis linking congestion patterns to emissions. First, it provides empirical evidence on the spatiotemporal and environmental dynamics of traffic congestion in Sweden. Second, it demonstrates a scalable and reproducible framework for transforming complex public traffic datasets into structured analytical workflows, thereby supporting strategic transport planning, sustainability assessment, and AI-augmented decision-making.

2. Research Questions

1. How can unstructured web-based traffic sensor data provided by Trafikverket be automatically extracted, structured, and stored to support reliable historical traffic analysis?

2. How can extracted historical traffic data be analyzed to identify spatial and temporal congestion hotspots, and how can these traffic states be linked to traffic-related emissions?

3. Literature Review

Sweden has an extensive network of traffic sensors deployed across highways and urban road segments. These sensors collect various types of traffic-related data, including vehicle counts, speed measurements, and vehicle classifications. However, direct extraction of raw data from physical sensor devices is often time-consuming, technically complex, and costly. In addition, sensor-generated data may contain missing values, noise, or inconsistencies (Hoogendoorn & Funk, 2017). Measurement errors can arise from device malfunction, maintenance issues, or adverse weather conditions, further affecting data reliability.

To mitigate these challenges, researchers and practitioners frequently rely on centralized platforms rather than accessing individual sensor devices directly. In Sweden, Trafikverket provides a reliable web-based platform that aggregates real-time traffic information from distributed sensor systems. Several open APIs are available for retrieving live transport data, including real-time locations of buses, trains, and ferries (The Swedish Transport Administration's open API for traffic information, 2026). However, no dedicated API exists for accessing structured historical traffic data. Instead, historical records are available only through the website interface, typically presented in tabular or graphical formats.

While numerous applications have been developed using Trafikverket's real-time APIs, comparatively limited research has focused on the systematic extraction and analysis of historical traffic data. This limitation is largely due to the technical complexity associated with extracting structured datasets from web interfaces. To address this gap, this study proposes a reproducible and automated data extraction pipeline designed to convert complex web-based information into standardized, machine-readable formats suitable for storage and analysis.

A robust web extraction system must be efficient, rerunnable, and resilient to operational errors. It should manage network interruptions, prevent excessive server requests, handle formatting inconsistencies, and eliminate duplicate database entries (Mitchell, 2018). Additionally, the pipeline must be capable of skipping failed extraction attempts while continuing processing of remaining sources and avoiding reprocessing previously collected records. The proposed framework follows established best practices in automated data acquisition to improve reliability, scalability, and data integrity.

The limited accessibility of structured congestion data has constrained comprehensive analyses of congestion patterns and their environmental implications. Trafikverket's sensor data are primarily used for real-time traffic updates and incident reporting. However, vehicle counts and speed measurements across different locations provide valuable indicators of congestion intensity and traffic dynamics. Such information can support a better understanding of spatial and temporal congestion patterns across Swedish cities.

Given Sweden's sustainability and climate objectives, analyzing when and where congestion occurs and assessing its associated emission impacts is critical for informed transportation planning and policy evaluation. Traffic volume and speed directly influence fuel consumption and CO₂ emissions, and variations in vehicle type further affect emission levels. By establishing a reliable historical data extraction pipeline, it becomes possible to analyze congestion patterns across time and space, thereby enabling integrated assessments of traffic performance and environmental impacts.

In the context of emissions and air quality, research has increasingly focused on quantifying the relationship between traffic patterns and pollutant levels. A study using traffic and air quality data in Dublin employed regression models to analyze how traffic dynamics influence urban pollution, demonstrating the importance of spatiotemporal information for emission modeling (Tafidis, Gholamnia, Sajadi, Vijayakrishnan, & Pilla, 2024). Similarly, high-resolution traffic and emission

inventories have been developed using big data approaches, revealing clear spatial and temporal variability in emission hotspots. These studies show that fine-grained datasets enable a more detailed understanding than traditional static inventories (Y. Wang et al., 2025). Research integrating dynamic traffic data for environmental monitoring highlights that real-time or high-resolution traffic information improves the fidelity of pollutant concentration estimates, emphasizing the value of rich traffic data in environmental modeling (Wen, et al., 2024).

Machine learning has become an increasingly dominant tool in traffic prediction and congestion detection. AI-powered systems that leverage spatiotemporal traffic data show improvements in short-term congestion prediction and forecasting accuracy, often outperforming classical regression methods and enabling more responsive traffic management strategies (Revolutionizing Short-term Traffic Congestion Prediction with Machine Learning, 2024). Such predictive capabilities are also critical for estimating operational metrics such as speed-flow relationships and congestion patterns, which in turn influence emission estimates and traffic signal timing optimization.

Recent developments in AI tool integration focus on enabling models to interact with external data systems and structured datasets in a standardized and maintainable way. A prominent example is the Model Context Protocol (MCP), an open-source integration standard designed to bridge AI applications with structured data sources. MCP establishes a client-server architecture in which AI models can query external datasets, execute analytical operations, and retrieve results without requiring direct database expertise (Introducing the next generation of Claude, 2024). This concept directly addresses practical challenges in transportation analytics by enabling large-scale traffic datasets to be analyzed efficiently and reproducibly.

Finally, research on AI-enhanced urban systems illustrates how integrated sensor networks, computer vision, and machine learning can contribute to traffic monitoring and emission estimation. Studies combining IoT sensors and AI-based traffic analysis demonstrate that traffic conditions can be directly linked to pollutant

trends, providing a foundation for adaptive traffic control and environmental assessment.

Traffic Congestion and Performance Measurement

Traffic congestion is a critical challenge in modern transportation systems, affecting mobility efficiency, travel reliability, and environmental sustainability. Congestion typically occurs when traffic demand approaches or exceeds roadway capacity, leading to reduced speeds, increased delays, and unstable traffic flow. Recent studies emphasize that congestion should be evaluated using multiple performance indicators rather than relying on a single metric, in order to capture its multidimensional nature (Afrin & Yodo, 2020).

Speed-Based Indicators and Speed–Flow Relationships

Average vehicle speed is a fundamental indicator of traffic performance and congestion. The relationship between speed, flow, and density is described by the fundamental diagram of traffic flow, which shows that vehicle speed decreases as traffic density increases beyond a critical threshold (Helbing, 2001). Recent empirical studies show that speed remains stable under free-flow conditions but declines rapidly under congested conditions (Han, et al., 2024).

Speed-based congestion metrics are particularly valuable because they directly reflect user experience. According to FHWA guidelines, speed reductions are among the most intuitive and policy-relevant indicators of congestion, translating directly into travel delays and reduced network reliability (Traffic Congestion and Reliability: Linking Solutions to Problems, 2017).

Delay Index and Travel Time Degradation

Traffic delay represents additional travel time caused by congestion relative to free-flow conditions. Delay-based metrics, such as the Delay Index (DI), quantify congestion in units directly interpretable by travelers, planners, and policymakers (Dixit, 2020). The Delay Index (DI) is defined as a normalized speed-based metric representing the proportional reduction in observed speed relative to free-flow conditions. This formulation differs from traditional travel-time-based delay

measures but provides a consistent and interpretable representation of congestion intensity when only speed data are available (Gaňan-Cardenas, Rios-Echeverri, Ballesteros, & Branch-Bedoya, 2024). Because the free-flow reference is estimated using the 95th percentile of observed speeds rather than a fixed upper bound, the metric may produce negative values in cases where traffic temporarily exceeds this statistical benchmark. These cases are retained in the analysis and interpreted accordingly.

Temporal Dynamics of Congestion

Congestion varies across hours, days, and seasons. Peak-hour congestion dominates total delay in urban and interurban networks (Xu, Sun, & Liu, 2024). Temporal disaggregation of traffic data enables identification of critical congestion windows and improves the assessment of reliability. The approach in this thesis aligns with best practices by evaluating congestion metrics over time intervals rather than relying on aggregated averages.

Vehicle-Type Heterogeneity

Traffic streams consist of multiple vehicle categories with distinct operational characteristics. Heavy vehicles, trailers, and multi-axle trucks generally exhibit lower acceleration and occupy more space, thereby amplifying congestion under high demand (Wang, Zhao, Ma, Wu, & Bao, 2023). Recent literature emphasizes that disaggregating analysis by vehicle category is essential for capturing true traffic dynamics. The framework in this thesis leverages vehicle-specific speed and flow data to account for heterogeneous traffic performance.

Integrated Congestion–Emission Analysis

Recent research increasingly supports integrated frameworks combining multiple congestion indicators with emission modeling. Emission estimation in road transport research is commonly based on speed-dependent emission factor models. Among these, the COPERT (Computer Programme to Calculate Emissions from Road Transport) framework is widely used in European studies to estimate vehicle emissions under varying traffic conditions. COPERT provides empirically derived

emission functions that capture the non-linear relationship between vehicle speed and pollutant output, making it particularly suitable for congestion-based emission analysis. In this study, the COPERT methodology is adopted to ensure consistency between the theoretical framework and the empirical analysis (European Environment Agency (EEA), 2019).

Vehicle category heterogeneity is especially important for emission estimates: heavy-duty vehicles contribute disproportionately to total emissions under congestion, while passenger cars exhibit different speed-emission profiles. High-resolution traffic datasets that differentiate vehicle types, such as those provided by Trafikverket's sensors, allow vehicle-specific emission estimation, aligning with the methodological focus of this thesis.

COPERT-based emission modeling represents a structured approach for linking congestion dynamics with environmental impacts. This framework addresses both methodological gaps (linking congestion and emissions using extracted web data) and practical gaps (making large-scale traffic datasets accessible and analyzable for researchers and policymakers).

4. Research Gap

Existing literature lacks practical and reproducible approaches that use observed traffic sensor data to:

- Identify spatial and temporal congestion hotspots based on measured speed and flow conditions; and
- Estimate traffic-related pollutant emissions by linking observed traffic states to speed-dependent emission factors.

Traffic congestion and transport-related emissions are closely interconnected challenges in modern transportation systems. Congestion affects mobility, travel reliability, and infrastructure efficiency, while road transport remains a significant contributor to air pollution and greenhouse gas emissions. Although both topics have been widely studied, they are often examined separately, particularly when using historical traffic sensor data.

While there is growing interest in frameworks that improve data accessibility and usability, evidence of fully integrated solutions remains limited. Existing approaches seldom demonstrate how large-scale traffic datasets can be queried and interpreted in practice. In this study, a Model Context Protocol (MCP) server was used primarily as a lightweight interface to simplify database access. The MCP server allowed interaction with structured datasets without requiring advanced SQL knowledge, but its role was restricted to basic usability support rather than offering any novel analytical or scientific contribution.

5. Motivation

This research is motivated by the need to better utilize Trafikverket’s existing traffic sensor infrastructure for integrated transport and environmental analysis. Given Sweden’s sustainability and climate objectives, understanding where and when congestion occurs and the associated emission implications is essential for informed planning and policy evaluation. By developing a framework that combines congestion hotspot detection, condition-sensitive emission estimation, and a user-friendly data access layer via the MCP server, this study enhances both the analytical and practical value of historical traffic data. The approach supports systematic, spatially explicit, and temporally sensitive assessment of traffic conditions and their environmental consequences, while demonstrating how AI integration standards can improve accessibility and reproducibility of transportation analytics.

6. Methodology

This thesis follows a quantitative, data-driven research methodology with an applied focus on business intelligence and data analytics. The objective is to design and implement an end-to-end data pipeline that collects, stores, analyzes, and reports traffic measurement data from the Swedish Transport Administration (Trafikverket). The methodology is structured into the following phases:

1. Manual input of the source URLs and data extraction
2. Automated data extraction, data parsing, validation, and storage

3. Centralized data management
4. Multi-tool analysis and reporting

This approach ensures reproducibility, scalability, and analytical reliability.

a. Technology Stack and Tools

This study employs a combination of open-source and industry-standard technologies selected for reliability, maintainability, and suitability for data-intensive analytical tasks.

Backend and Data Processing

Python (version 3.9.6 or later) was employed as the primary programming language for data extraction and processing. Selenium (version 4.0 or later) was utilized to automate browser interactions and scrape traffic data from the Trafikverket website, which contains dynamically rendered content. WebDriver Manager facilitated automated configuration and version management of ChromeDriver. Pandas was applied for data cleaning, transformation, and validation. PostgreSQL served as the central database for storing traffic data, with connectivity from Python managed through the psycopg2 adapter.

MCP Server and Reporting Layer

A Model Context Protocol (MCP) server is implemented to provide user-driven and programmatic access to the stored traffic data. The server is developed using TypeScript (version 5.3 or later) and runs on Node.js. It uses the modelcontextprotocol/sdk for protocol implementation and the pg library for PostgreSQL connectivity. Environment configuration is managed using dotenv, and ts-node is used during development to execute TypeScript code.

Development and Automation Tools

To support automation and cross-platform execution, both Windows batch scripts (RUN.bat, SETUP-WIZARD.bat, INSTALL.bat) and shell scripts (run.sh, setup.sh) are used. Git is used for version control to ensure traceability and reproducibility.

b. Data Input and Extraction

The data extraction process begins with manual user input, where the user provides one or more URLs from the Trafikverket traffic information system:

<https://vtf.trafikverket.se/SeTrafikinformation>

These URLs define the scope of traffic data to be collected. Once provided, the Python-based scraper uses Selenium and a headless Chrome/Chromium browser to retrieve traffic measurement data. The extraction process is designed to be rerunnable. Before inserting any record into the database, the system checks whether the same data identified by measurement time, location (county, road number, and measurement point), and vehicle metrics already exists. If a duplicate record is found, insertion is skipped. This ensures that repeated executions do not introduce redundant data.

c. Data Parsing and Storage

Trafikverket's interface is presented in Swedish; therefore, translation and normalization processes are incorporated into the data preprocessing workflow. After extraction, the raw traffic data are parsed and transformed into a structured format suitable for analytical use. Validation rules are applied to ensure data consistency, correct data types, and completeness. The processed data are stored in a PostgreSQL database (traffic_data), which acts as the single source of truth for all analytical tools. The database schema is designed to support analytical queries and includes:

- Location attributes (county, road number, measurement point)
- Measurement timestamps
- Vehicle counts and average speeds across multiple vehicle categories
- Metadata fields for audit and traceability
- Indexes on frequently queried fields such as measurement time, road number, and county to improve query performance.

d. System Architecture Overview

The complete data flow from manual input to analysis and reporting is illustrated in Figure 1.

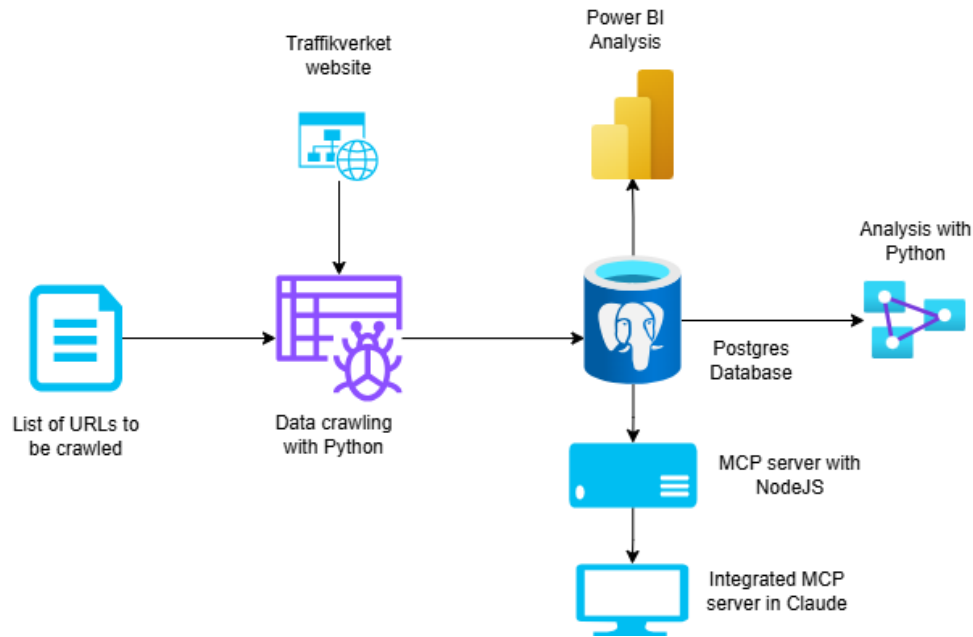


Figure 1: System architecture for data extraction to analysis

The figure shows how user-provided URLs are used as input for Python-based data crawling from the Trafikverket website. The extracted data are stored in a PostgreSQL database which serves as the data source for Power BI dashboards, Python-based analysis and MCP (Model context protocol) server. All analytical tools retrieve data from the database using structured SQL queries. In the Python-based analytical workflow, the complete dataset is extracted and processed to perform statistical and exploratory analyses.

In contrast, the MCP server enables more targeted data retrieval by executing parameterized queries based on specific attributes such as location, vehicle type, and time interval. This structured query mechanism facilitates efficient interaction between the database and AI-based analytical tools. In this implementation, the MCP server acts as an intermediary layer that allows an AI application (Claude Desktop) to query the database dynamically and generate analytical reports based on the stored traffic data.

7. Data

a. Data Source and Scope

The primary data source for this research is the **Swedish Transport Administration (Trafikverket)** traffic information system. The dataset consists of historical traffic measurements collected via roadside sensors and monitoring stations. While the Trafikverket network provides broad national coverage, the specific analysis in this thesis is focused on a selected set of measurement points across four Swedish counties. The cleaned and formatted data were stored in the database using the data extraction tool. The final dataset contained **5,137 records**, capturing longitudinal traffic behavior through hourly observations. The initial raw dataset contained 5,641 observations collected from the Trafikverket platform. Following data cleaning and validation procedures, including the removal of incomplete records, parsing inconsistencies, and duplicate entries, the final analytical dataset was reduced to 5,137 observations. All subsequent analyses in this study were conducted using this validated dataset.

b. Data Characteristics and Variables

The data retrieved from Trafikverket consist of semi-structured tabular information. Each record represents an hourly interval of traffic activity and contains the following key attributes:

Temporal Information: Measurement timestamps recorded at hourly intervals.

Spatial Information: Geographic metadata including the County (Län), Road Number (Vägnr), and the specific Measurement Point ID (Punktnummer).

Traffic Flow (Volume): Quantitative counts of vehicles passing the sensor, disaggregated by vehicle type.

Average Speed: Mean speed recorded in km/h for the corresponding time interval and vehicle category.

Vehicle Classification: Differentiated data across multiple categories, including passenger cars (with and without trailers), heavy vehicles, and various tractor-trailer configurations.

c. Database Schema and Structure

The following table outlines the structure of the traffic_data table within the PostgreSQL database. This schema is designed to support high-resolution historical analysis and AI-driven querying.

Column Name	Data Type	Description
id	SERIAL PRIMARY KEY	Unique identifier for each record.
measurement_time	TIMESTAMP	The date and hourly interval of the sensor observation.
county	VARCHAR(100)	The Swedish region (Län) where the sensor is located.
road_number	VARCHAR(10)	The specific road designation (Vägnr).
punkt_nummer	VARCHAR(20)	The unique ID for the measurement station.
all_vehicles_count	INTEGER	Total count of all vehicles for that hour.
all_vehicles_avg_speed	DECIMAL(5, 2)	Mean speed (km/h) for all vehicles.
passenger_car_count	INTEGER	Specifically recorded passenger car volume.
passenger_car_avg_speed	DECIMAL(5, 2)	Mean speed (km/h) for passenger cars.

heavy_vehicles_count	INTEGER	Specifically recorded heavy vehicle volume.
heavy_vehicles_avg_speed	DECIMAL(5, 2)	Mean speed (km/h) for heavy vehicles.
*_trailer_count	INTEGER	Breakdown columns for vehicles with/without trailers.
created_at	TIMESTAMP	Audit timestamp for record insertion.

Table 1: Traffic data table structure in database

d. Data Validation

To validate the data extracted using the tool, the following procedure was applied. First, an input URL (<https://vtf.trafikverket.se/tmg101/AGS/tmg104bestaellinfouttag.aspx?punktnrlista=13520474,13520475,13520478&laenkrollista=1,1,1>) was selected and opened in a web browser. The browser displayed 13 date ranges associated with the same Punktnummer. In the pop-up window, all available checkboxes were selected, and the Tabular option was chosen to present the data. Once the table was generated, it was copied and pasted into an Excel file. For each subsequent date range, the same procedure was repeated, and the resulting data was appended to the same Excel file.

After collecting all data for the given punktnummer, the same URL was processed using the scraper. Once the scraping was completed, the database table was queried for the same punktnummer, and the results were exported as a CSV file. The Excel and CSV outputs were then compared. The comparison yielded a 100% match in traffic values and timestamps across all tested ranges. The results for the 13 validated date ranges are summarized below:

Date Range (Start - End)	Expected Rows (Manual)	Actual Rows (Scraped)	Result
2023-09-05 to 2023-09-06	24	24	Match
2023-06-12 to 2023-06-13	24	24	Match
2023-06-09 to 2023-06-12	72	72	Match
2023-03-20 to 2023-03-21	24	24	Match
2021-10-11 to 2021-10-12	0	0	Match (Empty)
2021-07-15 to 2021-07-16	24	24	Match
2021-04-13 to 2021-04-14	24	24	Match
2019-03-18 to 2019-03-19	0	0	Match (Empty)
2015-10-12 to 2015-10-13	24	24	Match
2015-08-07 to 2015-08-10	72	72	Match
2015-08-06 to 2015-08-07	24	24	Match
2015-03-23 to 2015-03-24	0	0	Match (Empty)
2015-03-20 to 2015-03-23	72	72	Match

Table 2: Traffic data row count by date range

Out of the 13 checked ranges, 3 (approximately 23%) contained no data on the source website. The scraper correctly identified these empty states and did not record any entries, confirming that the tool does not generate "ghost" data or fail when encountering empty data. No mismatches or parsing failures were encountered during this audit. The only differences observed between the source and the database were the addition of computed metadata (e.g., `created_at` timestamps) and normalized columns (`county`, `road_number`), which are intended features of the extraction pipeline.

The following data integrity measures are implemented within the Python-based extraction pipeline:

Composite Key Duplicate Detection: The system utilizes a composite unique constraint on (`measurement_time`, `county`, `road_number`, `punkt_nummer`). Before any record is inserted, the system verifies whether the data already exists, ensuring that repeated executions do not introduce redundant information.

Type Parsing and Cleaning: Raw strings extracted from the web interface are cleaned using the **Pandas** library, converting them into validated numeric types (integers and decimals), thereby reducing the risk of analytical errors.

Auditable Extraction: The use of **Selenium** enables the pipeline to interact with the Trafikverket user interface in a manner that mimics manual selection, including choosing the "Table" format and the relevant vehicle metrics. This approach supports consistent data extraction aligned with the source pages. The observed accuracy of the extracted values within the tested ranges is supported through manual auditing of selected outputs rather than being solely guaranteed by the automated interaction.

Centralized Storage: All data are maintained in a PostgreSQL database, which serves as the single source of truth for Power BI, Python statistical modules, and the MCP server.

e. Data Analysis and Visualization

The PostgreSQL database supports three analytical components:

1. Power BI is used for interactive dashboards and visual exploration of traffic patterns, including temporal trends, vehicle-type distributions, and regional comparisons.
2. Python with Pandas is used for exploratory data analysis, statistical summaries, and custom visualizations.
3. The MCP Server provides user-driven analytical reports based on structured or natural-language input, abstracting database complexity from end users.

f. Data Integrity and Reliability

The reliability of the dataset was evaluated through a series of functional tests designed to verify the consistency of the extraction pipeline and the prevention of data redundancy. These mechanisms include:

- **Automated Duplicate Detection:** Logic implemented during the database insertion phase to prevent record duplication.
- **Idempotent Extraction:** Logic ensuring that the tool is rerunnable without altering the existing state of the database.
- **Centralized Storage:** Use of a PostgreSQL database to maintain a single source of truth for subsequent analysis.

The tool was initially executed using two input URLs specified in the `input_url` configuration file. Following execution, the database was inspected to verify that the extracted records had been correctly inserted and properly formatted. The tool was subsequently run again using the same set of URLs. During this second execution, no additional rows were inserted into the database, suggesting that the system can prevent duplicate entries.

To further evaluate the incremental extraction functionality, a new URL was appended to the `input_url` file, and the tool was executed once more. In this case, only the records corresponding to the newly added URL were inserted into the database, demonstrating that the pipeline can correctly identify and process previously unextracted sources.

Finally, all existing URLs in the configuration file were commented out using the `#` prefix, and a single new URL was added without being commented. Upon execution, the system processed only the uncommented URL, indicating that the crawler selectively executes based on active (non-commented) entries.

A comprehensive step-by-step guide detailing the setup and execution procedures is provided in the Appendix.

g. Ethical and Legal Considerations

The dataset contains no personally identifiable information. All traffic data are aggregated and publicly available through Trafikverket, ensuring compliance with ethical research standards and data protection regulations.

h. Methodology Summary

This methodology presents a complete and scalable analytical pipeline for traffic data analysis. By combining automated web scraping, structured database storage, and multi-platform analytical tools, the study demonstrates how modern business intelligence techniques can support data-driven decision-making in traffic analysis.

8. Data Analysis

This chapter presents a comprehensive analysis of traffic congestion and its environmental impact on Swedish road networks. The analysis is structured in three main parts: (1) traffic congestion indicator validation and methodology, (2) spatiotemporal congestion patterns, and (3) the relationship between traffic congestion and CO₂ emissions. The study utilizes 5,137 observations collected from eight measurement points across four counties (Dalarnas län, Stockholms län, Uppsala län, and Kronobergs län) spanning from January 1998 to November 2023.

a. Traffic Congestion Indicators: Methodology and Validation

The foundation of congestion analysis relies on accurately determining free-flow speed conditions and establishing robust indicators. This section details the methodological framework and validation of three primary congestion metrics: Speed Ratio (SR), Delay Index (DI), and Speed Variability.

Free-Flow Speed Determination

Free-flow speed (v_{ff}) represents the speed at which vehicles travel under ideal conditions with minimal traffic interference. Following the methodology established by the Highway Capacity Manual (HCM, 2016), the 95th percentile of observed speeds was selected as the most robust estimator of free-flow conditions, as it effectively filters out extreme outliers while capturing the upper range of unimpeded travel speeds.

The 95th percentile speed is calculated as:

$$V_{ff} = P_{95}(V_{obs})$$

Where $P_{95} (V_{obs})$ denotes the 95th percentile of all observed speeds in the dataset. Table 3 presents the calculated free-flow speeds for different vehicle categories. The analysis reveals distinct speed characteristics across vehicle types, with passenger cars achieving the highest free-flow speeds (92.8 km/h) and heavy vehicles with trailers showing the lowest (85.0 km/h). This differentiation is critical for accurate congestion assessment, as it reflects the operational constraints and performance capabilities of different vehicle classes.

Free-Flow Speed Statistics by Vehicle Category

Vehicle Category	Maximum (km/h)	95th Pct (km/h)	Mean (km/h)	Median (km/h)
All Vehicles	139.70	91.90	70.77	69.30
Passenger Cars	142.40	92.80	71.46	69.85
Heavy Vehicles	112.00	87.70	67.05	65.80
Passenger Car + Trailer	110.90	85.20	64.75	63.50
Heavy Vehicle + Trailer	99.00	85.00	64.81	62.90

Table 23: Free-flow speed statistics by vehicle category

Note: Bold values indicate the 95th percentile speeds used for free-flow reference.

Speed Ratio (SR)

Speed Ratio serves as the primary congestion indicator, representing the proportion of free-flow speed achieved under observed traffic conditions. The Speed Ratio is calculated as:

$$SR = V_{obs} / V_{ff}$$

where V_{obs} is the observed average speed during the measurement period, and V_{ff}

is the free-flow speed determined from the above equation. Values of SR closer to 1.0 indicate conditions approaching free-flow, while lower values represent increasing levels of congestion.

Based on the Speed Ratio values, traffic conditions are classified into four distinct congestion levels:

- Free-flow: $SR \geq 1.0$ (speeds equal to or exceeding free-flow conditions)
- Light congestion: $0.8 \leq SR < 1.0$ (minor speed reductions)
- Moderate congestion: $0.6 \leq SR < 0.8$ (noticeable speed reductions)
- Severe congestion: $SR < 0.6$ (significant speed degradation)

The analysis of 5,137 valid observations reveals the following distribution across congestion levels: severe congestion accounts for 10.9%, moderate congestion dominates with 53.8%, light congestion represents 30.3%, and free-flow conditions occur in only 5.1% of cases. The mean Speed Ratio of 0.770 indicates that, on average, traffic operates at approximately 77% of free-flow speed, suggesting persistent moderate congestion across the study area.

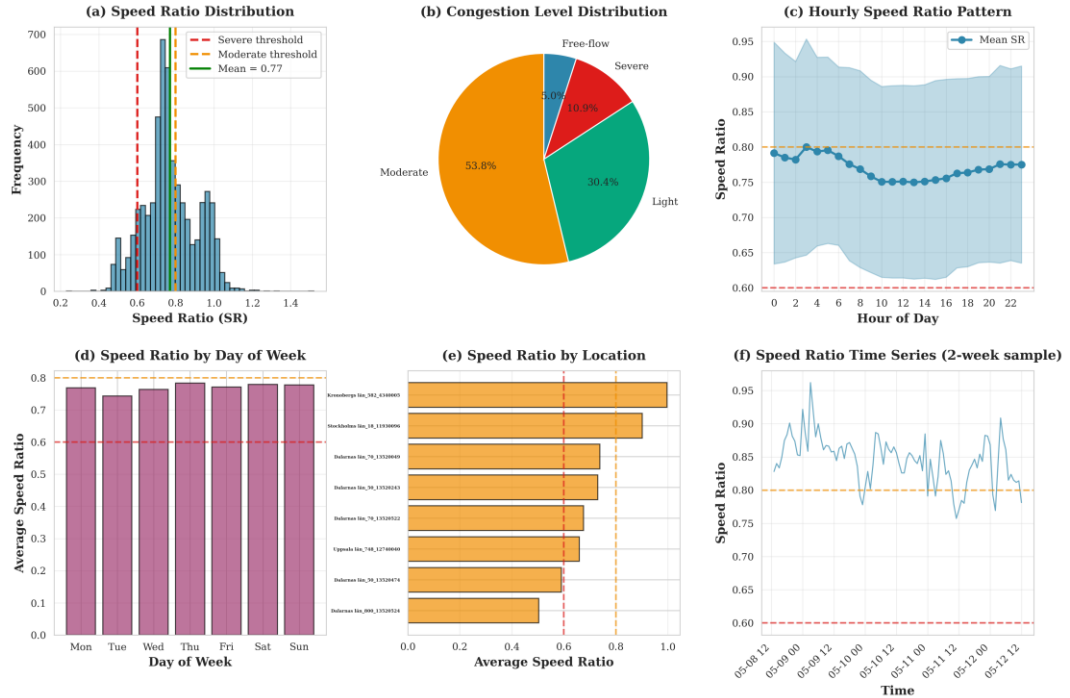


Figure 2: Speed ratio analysis

Delay Index (DI)

The Delay Index quantifies the relative time loss experienced by travelers due to congestion. Following the normalized formulation recommended by the Federal Highway Administration (FHWA),

the Delay Index is calculated as:

$$DI = 1 - (V_{obs}/V_{ff})$$

This formulation is selected due to the absence of direct travel time measurements in the dataset. Since the available data consists of aggregated speed observations, a speed-based representation of delay ensures methodological consistency across all analyses.

This formulation produces values primarily within the range of 0 to 1 under typical traffic conditions, where $DI = 0$ represents free-flow conditions (no delay) and $DI = 1$ indicates a complete standstill. However, negative values may occur when the observed speeds exceed the estimated free-flow speed. A Delay Index of 0.3 indicates that average speed is reduced by 30% relative to free-flow conditions, meaning that vehicles operate at 70% of the free-flow speed. Because travel time is inversely proportional to speed, this corresponds to an approximately 43% increase in travel time compared to free-flow conditions.

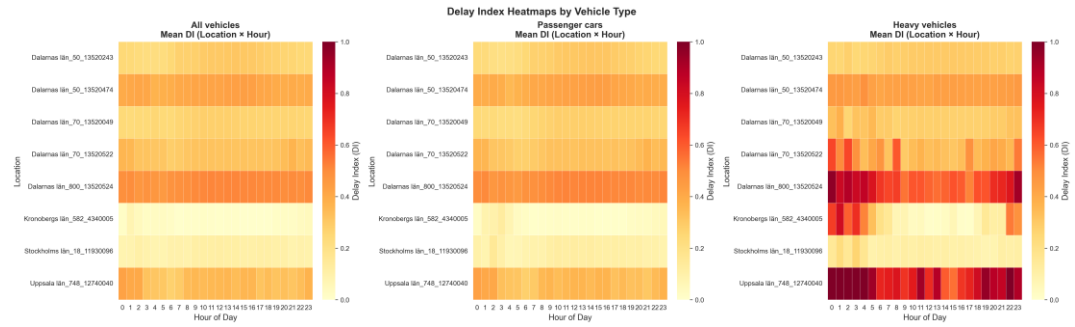


Figure 3: Delay index heatmap by vehicle type

The analysis reveals Delay Index values ranging from -0.520 to 0.769, with a mean of 0.230 and a median of 0.246. Negative DI values arise when observed speeds exceed the estimated free-flow reference ($SR > 1$), reflecting instances in which traffic conditions temporarily surpass the statistical free-flow benchmark. Such observations were retained in the dataset to preserve analytical completeness and to avoid potential bias introduced by truncation. This behavior highlights an inherent

limitation of percentile-based free-flow estimation, where the chosen reference value does not represent a strict upper bound but rather a high-probability operating condition. The mean DI of 0.230 indicates that, on average, speeds are reduced by 23% relative to free-flow conditions, corresponding to the observed mean Speed Ratio of 0.770. This speed reduction implies an approximate 30% increase in travel time compared to ideal conditions.

Speed Variability

Speed variability provides a measure of traffic flow instability within specified time windows. The coefficient of variation (CV) of speeds is calculated for each hourly observation window as:

$$CV = \sigma_v / \mu_v$$

where σ_v is the standard deviation of speeds and μ_v is the mean speed within the time window. Higher CV values indicate unstable traffic flow, often associated with stop-and-go conditions characteristic of severe congestion. A threshold of $CV > 0.3$ is typically used to identify periods of high variability.

The speed variability metric complements SR and DI by capturing the quality of traffic flow beyond average speed conditions. While two time periods may exhibit identical average speeds, high variability in one period indicates unstable, uncomfortable conditions with frequent acceleration and deceleration cycles, which further impacts fuel consumption and emissions.

b. Spatiotemporal Patterns of Traffic Congestion

Understanding the spatial and temporal distribution of traffic congestion is essential for targeted mitigation strategies. This section presents a comprehensive analysis of congestion patterns across different measurement locations and time periods, with differentiation between passenger cars and heavy vehicles.

Spatial Distribution of Congestion

The spatial analysis aggregates congestion indicators by location, identifying hotspots with consistently poor traffic conditions.

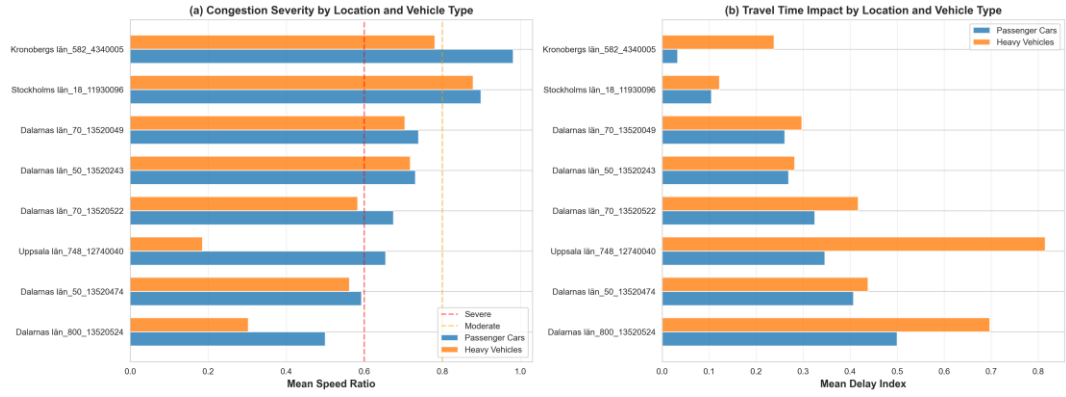


Figure 4: Spatial hotspot by vehicle type

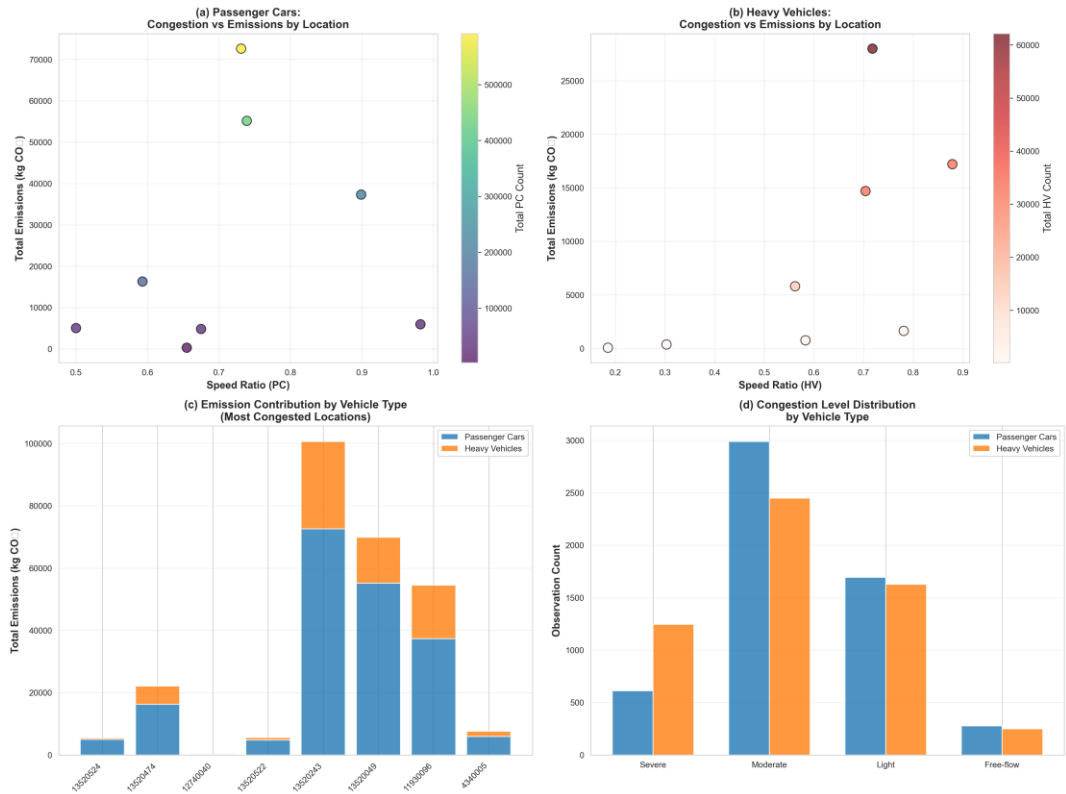


Figure 5: Integrated congestion emission analysis

Figure 4 presents the spatial distribution of average Speed Ratio and Delay Index across the eight measurement points, while Figure 5 displays the corresponding CO₂ emissions by location, revealing significant inter-location variability in both congestion severity and environmental impact.

The spatial distribution suggests a five-fold performance difference between best and worst locations. Kronobergs län (Road 582) and Stockholms län (Road 18) maintain near-free-flow conditions ($SR > 0.85$ for passenger cars), while Uppsala län (Road 748) and Dalarnas län (Road 800) represent critical bottlenecks ($SR <$

0.60). Emission analysis reveals that total CO₂ burden depends on both congestion severity and traffic volume-the highest-emission location generates ~73,000 kg despite excellent Speed Ratio due to high traffic volume, while severely congested locations with lower volumes produce fewer total emissions.

These spatial hotspots show consistent patterns across both vehicle types, though heavy vehicles experience systematically more severe congestion at all locations (8-12% lower Speed Ratios on average). This pattern may be related to operational characteristics of heavy vehicles, such as longer acceleration times and reduced maneuverability, although these factors are not directly observed in the dataset. Heavy vehicles account for approximately 25-30% of total emissions in the dataset, despite representing a smaller share of traffic volume. This suggests a disproportionate contribution to overall emissions.

Road characteristics may influence congestion patterns. Locations on primary highways (Road 70 and Road 50) generally show better Speed Ratios compared to secondary roads (Road 800, Road 748), which exhibit the most severe congestion. The observed heavy-vehicle disadvantage appears to persist across all road types, with heavy vehicles experiencing severe congestion ($SR < 0.6$) approximately 4× more frequently than passenger cars.

c. Temporal Patterns and Peak Hour Analysis

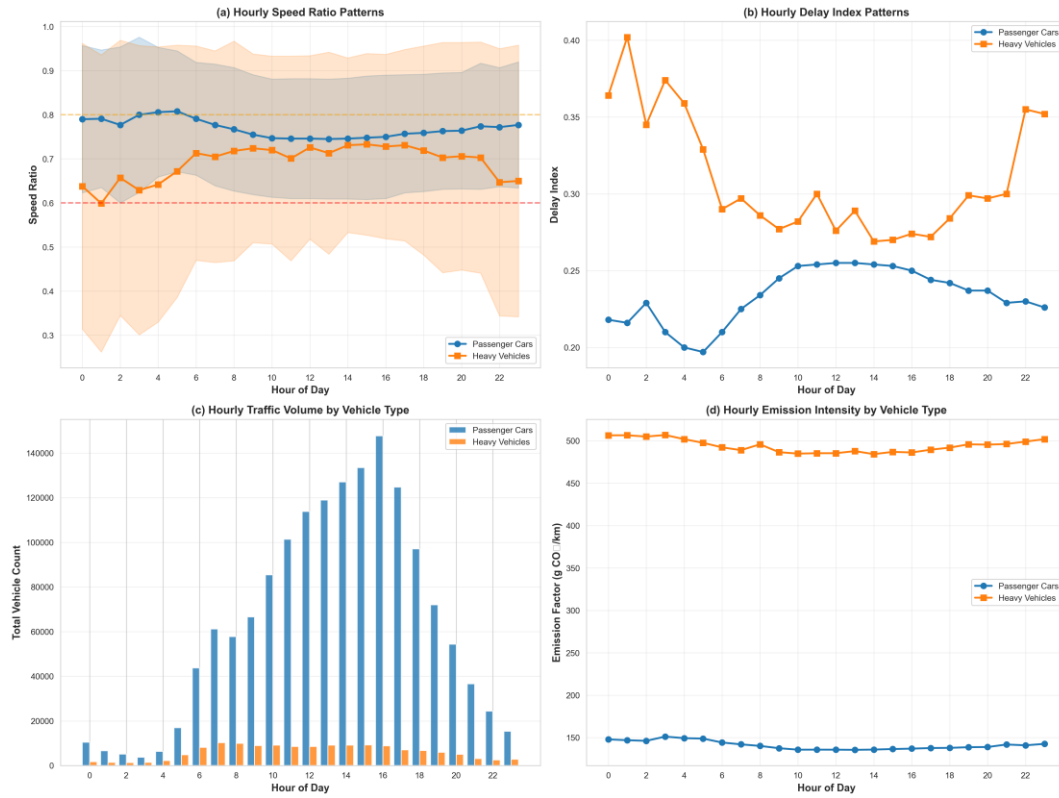


Figure 6: Hourly temporal pattern

Temporal analysis reveals distinct patterns in congestion intensity throughout the day. Figure 6 illustrates the hourly variation in mean Speed Ratio for both passenger cars and heavy vehicles across all measurement locations.

For passenger cars, the temporal profile is notably stable throughout the day, contrasting with typical urban commuting patterns. Mean Speed Ratios remain relatively constant around 0.78-0.80 across most hours, with only minor fluctuations. The slight dip during morning hours (7:00-9:00 AM) shows SR values around 0.77, while midday and afternoon periods (10:00 AM - 6:00 PM) maintain SR values of approximately 0.76-0.78. This limited temporal variation (total range of only 0.04-0.05 SR points across the entire day) suggests that passenger car congestion on these predominantly rural and primary highway corridors may be influenced more by location-specific capacity constraints than by strongly peaked commuter demand. Heavy vehicles exhibit a markedly different temporal profile with substantially greater hour-to-hour variation. The most pronounced feature is severe early-morning degradation (hours 1:00-5:00 AM) where mean SR drops to 0.62-0.65, representing moderate to severe congestion during a period when overall

traffic volumes are minimal. This pattern may reflect operational constraints, regulatory speed restrictions, or routing behavior; however, these factors are not directly captured in the available data. Heavy vehicle conditions improve during morning hours (6:00 AM- 9:00 AM), reaching peak performance during midday business hours (10:00 AM - 6:00 PM), with SR values of 0.71-0.73. Evening hours (7:00 PM - midnight) show a gradual deterioration back to SR values of 0.65-0.67.

Night-time and early morning hours (10:00 PM - 5:00 AM) show contrasting patterns between vehicle types: passenger cars maintain relatively good Speed Ratios (0.79-0.80), approaching their best daily performance, while heavy vehicles experience their worst conditions during hours 1:00-5:00 AM (SR = 0.62-0.65). This 0.15-0.18-point gap during overnight hours represents the maximum performance disparity between vehicle types across the diurnal cycle.

The amplitude of temporal variation differs substantially by vehicle type. Passenger cars exhibit approximately 5-6 km/h variation in mean speeds across the 24-hour cycle (69-75 km/h), while heavy vehicles show approximately 12 km/h variation (52-64 km/h)-twice the temporal sensitivity of passenger cars. This suggests that heavy vehicle operations may be more susceptible to time-of-day factors beyond simple traffic volume, including driver scheduling practices, delivery time windows, and potential nighttime operational restrictions.

Importantly, pronounced morning and evening peak-hour effects characteristic of urban commuter corridors are not strongly evident in the aggregate passenger car data. This is consistent with these measurement locations function primarily as inter-regional highways and rural arterials rather than urban commuter routes. The persistent moderate congestion levels (mean SR = 0.77) across most hours suggest the presence of relatively persistent capacity constraints or high ambient traffic levels rather than peaked demand patterns that could be addressed through traditional peak-hour management strategies. However, temporal variation is more noticeable for heavy vehicles, indicating that peak-hour effects may be more relevant for specific vehicle categories rather than uniformly across all traffic.

d. Vehicle Type Differentiation in Congestion Experience

A critical finding of this analysis is the systematic difference in congestion experience between passenger cars and heavy vehicles. Across all locations and time periods, heavy vehicles consistently exhibit lower Speed Ratios than passenger cars, indicating more severe congestion impact.

Passenger cars appear to maintain higher speed ratios under similar conditions, while heavy vehicles exhibit earlier speed reductions. This observed difference may reflect underlying operational characteristics, although these are not explicitly measured in the dataset. On average, heavy vehicle Speed Ratios are 8-12% lower than passenger car Speed Ratios at the same location and time.

This differential impact may have implications for both traffic management and environmental analysis. The observed patterns suggest that freight-related congestion may not be fully reflected when using aggregate traffic measures alone. These findings may warrant further investigation into vehicle-type-specific traffic management strategies, such as truck-specific lanes or time-of-day restrictions. However, the present study does not directly evaluate the effectiveness of such interventions.

e. Congestion-Emission Relationship Analysis

This section examines the quantitative relationship between traffic congestion and CO₂ emissions, a critical link for understanding the environmental consequences of traffic management policies.

CO₂ Emission Calculation Methodology

CO₂ emissions are calculated using speed-dependent emission factors derived from the COPERT (Computer Programme to Calculate Emissions from Road Transport) methodology, which accounts for the non-linear relationship between vehicle speed and fuel consumption.

The emission calculation for each observation is:

$$E = \sum_i (N_i \times EF_i(v) \times d)$$

where E is total CO₂ emissions (grams), N_i is the count of vehicles in category i , $EF_i(v)$ is the speed-dependent emission factor for vehicle category i (g/km), v is the observed average speed, and d is the distance traveled (assumed as unit length for comparative analysis).

The emission factors follow a U-shaped curve with respect to speed, with minimum emissions occurring at optimal speeds (typically 60-80 km/h for passenger cars, 50-70 km/h for heavy vehicles). Both very low speeds (stop-and-go congestion) and very high speeds result in elevated emissions per kilometer traveled.

Aggregate Emission Patterns

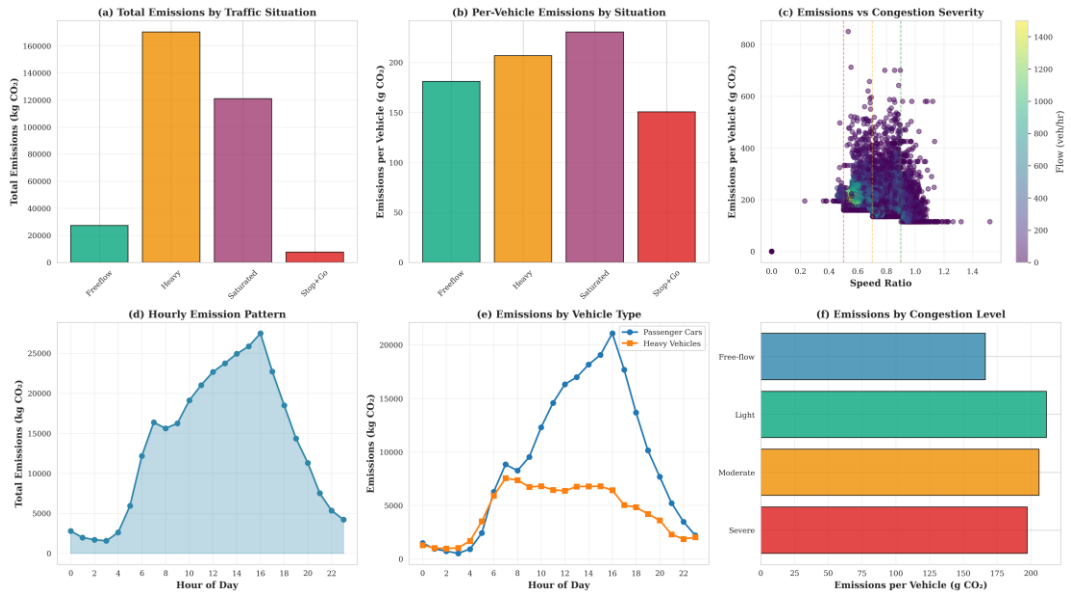


Figure 7: Integrated emission analysis (across the study period)

The total CO₂ emissions across all observations in the study period provide insight into the environmental burden of traffic congestion. Figure 7 reveals that heavy traffic conditions (moderate congestion) generate the highest total emissions (~165,000 kg CO₂) across the study period, substantially exceeding free flow (~27,000 kg), saturated (~120,000 kg), and stop-go conditions (~7,000 kg) (panel a). This pattern reflects the combined effect of high traffic volumes operating under moderately reduced speed conditions that are both common and emissions intensive.

Temporal patterns in emissions exhibit clear diurnal variation, with peak emission hours occurring during afternoon traffic periods (panel d). Total emissions reach maximum levels around hour 16-17 (~27,000 kg), declining progressively through evening hours and reaching overnight minimums of ~2,000 kg during hours 2-4. The hourly emission pattern demonstrates a strong correlation with traffic volume patterns, confirming that absolute emissions are driven primarily by the number of vehicles on the road rather than congestion severity alone.

The relationship between congestion and per-vehicle emissions is non-linear (panel c). The scatter plot of Speed Ratio versus emissions per vehicle reveals that as Speed Ratio decreases below 0.8, emission rates increase substantially. Per-vehicle emissions range from ~180 g CO₂/vehicle during free-flow conditions to ~230 g CO₂/vehicle under saturated conditions (panel b) representing approximately a 28% increase. However, the most emissions-intensive condition is saturated flow rather than stop-go traffic, as saturated conditions combine moderately elevated emission factors with high traffic volumes, while stop-go conditions involve fewer total vehicles despite potentially higher per-vehicle rates during acceleration events.

Heavy vehicles contribute disproportionately to total emissions throughout the day (panel e). The hourly emission pattern reveals distinct temporal profiles by vehicle type: heavy vehicles generate peak emissions of ~7,500 kg during morning hours (hour 7), aligning with freight delivery schedules, while passenger cars peak at ~21,000 kg during afternoon hours (hour 16), reflecting commuter travel patterns. At the afternoon peak, passenger cars account for approximately 74% of total emissions (21,000 kg) while heavy vehicles contribute 26% (~7,000 kg). Given that heavy vehicles typically represent only 10-15% of traffic volume, their 26% emission contribution represents a disproportionate environmental impact. This disparity may be associated with several factors, including inherently higher emission factors for heavy vehicles, observed differences in congestion patterns between vehicle types, and the sustained presence of heavy vehicle traffic throughout business hours rather than concentrated peak periods. However, these factors were not directly measured in the dataset and should therefore be interpreted cautiously.

The distribution of emissions by congestion level (panel f) demonstrates that moderate congestion conditions produce the most concentrated and substantial per-vehicle emission impacts. While severe congestion generates elevated per-vehicle emissions, the relatively rare occurrence of truly severe conditions ($SR < 0.6$) limits their contribution to total emission burden. Conversely, moderate congestion ($SR = 0.6-0.8$), which accounts for 53.8% of all observations, represents the dominant emission regime. This finding carries important policy implications, suggesting that interventions targeting moderate congestion conditions may yield greater aggregate emission reductions than strategies focused solely on severe congestion hotspots, despite the latter exhibiting higher per-vehicle emission intensities.

Speed-Emission Relationship

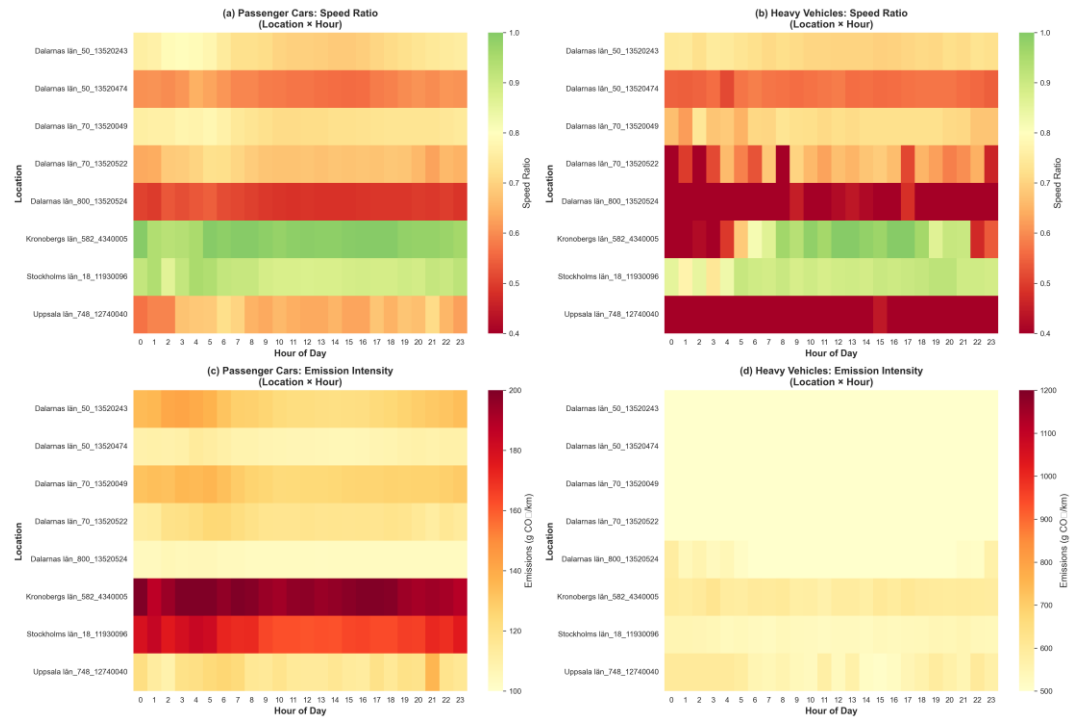


Figure 8: Spatial temporal heatmap

Figure 8 illustrates the relationship between Speed Ratio and per-vehicle CO₂ emissions. The results indicate a non-linear association, where lower Speed Ratio values (representing more severe congestion) are generally associated with higher emission rates.

This pattern is consistent with established findings in the literature, where both very low speeds and unstable traffic conditions are linked to increased fuel consumption and emissions. However, no formal model fitting was performed in this study, and the relationship should therefore be interpreted as descriptive rather than predictive.

Congestion-Emission Hotspots

Integrating spatial congestion patterns with emission calculations identifies priority locations for intervention. Figure 8 presents spatiotemporal heatmaps showing the joint distribution of congestion severity and emission intensity across locations and hours of the day.

Three distinct hotspot types emerge from the analysis:

Chronic congestion hotspots: Locations with persistently low Speed Ratios (mean $SR < 0.65$) throughout most daylight hours. These sites generate continuously elevated emissions and represent priority intervention targets.

Peak-hour hotspots: Locations that transition from free-flow or light congestion to severe congestion during rush hours. While the cumulative emission burden is lower than chronic hotspots, the sharp degradation during peaks suggests capacity constraints.

Volume-driven hotspots: Locations with moderate congestion but very high traffic volumes, resulting in substantial absolute emissions despite relatively better Speed Ratios. These sites may benefit from demand management rather than capacity expansion.

The geographic clustering of hotspots along primary corridors suggests that congestion and emission problems propagate spatially. Interventions at bottleneck locations may therefore yield benefits extending beyond the immediate intervention site.

f. Flow-Speed Regime Analysis

The relationship between traffic flow and speed provides insight into road capacity utilization and the transition between different traffic regimes. Flow-speed analysis complements traditional congestion indicators by revealing the mechanisms underlying observed congestion patterns.

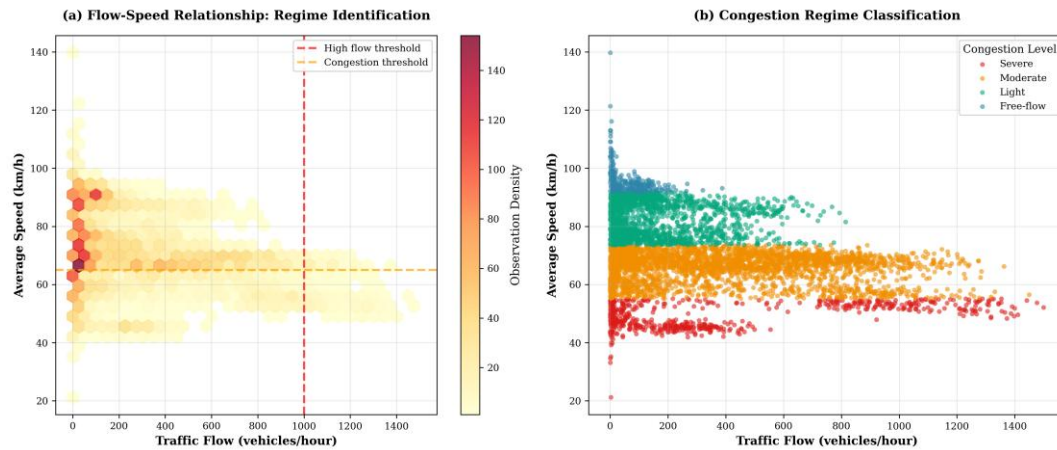


Figure 9: Speed flow regimes

Figure 9 (diagnostic plots) presents the flow-speed scatter for all observations, revealing three distinct regimes consistent with fundamental traffic flow theory: Uncongested regime: At low to moderate flows (< 800 - 1000 vehicles/hour), speeds remain near free-flow levels, showing minimal correlation with traffic volume. This regime represents under-capacity conditions where the road can accommodate additional demand without speed degradation.

Transitional regime: At flows approaching capacity (1000 - 1300 vehicles/hour), speeds begin declining as interactions between vehicles increase. This regime exhibits high variability, with small flow changes producing large speed variations, characteristic of unstable traffic conditions.

Congested regime: At high flows or when capacity is exceeded, speeds drop sharply and remain low despite flow reductions. This is consistent with breakdown conditions where the road operates in forced flow, with density rather than demand controlling speed. The observed flow-speed patterns show considerable scatter, reflecting the influence of factors beyond simple volume-capacity relationships, including vehicle mix, weather conditions, incidents, and driver behavior. The

maximum observed flow of 1,500 vehicles/hour provides an empirical estimate of practical capacity, though true capacity may be higher under ideal conditions.

The data suggests that heavy vehicles may have a notable influence on flow-speed relationships, as segments with higher heavy vehicle percentages are generally associated with lower speeds at equivalent flow rates. While the observed patterns indicate that heavy vehicles may affect traffic performance differently from passenger cars. Further targeted analysis would therefore be required to quantify such effects more precisely.

g. Summary of Key Findings

The comprehensive analysis of 5,137 observations across Swedish road networks reveals several critical findings regarding traffic congestion and its environmental impacts:

1. Persistent moderate congestion: The mean Speed Ratio of 0.770 indicates that traffic operates at approximately 77% of free-flow conditions on average, with 64.7% of observations showing moderate to severe congestion ($SR < 0.8$). This represents a significant mobility and environmental challenge.
2. Vehicle type disparities: Heavy vehicles consistently experience 8-12% lower Speed Ratios than passenger cars at the same locations and times, indicating differential congestion impacts that must be considered in traffic management and environmental assessment.
3. Non-linear congestion-emission relationship: CO₂ emissions appear to increase non-linearly as congestion worsens, with severe congestion ($SR < 0.6$) producing 40-60% higher per-vehicle emission rates compared to free-flow conditions. This suggests that reducing severe congestion may yield disproportionate environmental benefits.
4. Spatiotemporal concentration: Congestion and emissions exhibit strong spatial clustering, with a small number of hotspot locations accounting for a disproportionate share of total delay and environmental impact. These hotspots persist across time, which may indicate underlying capacity or design-related constraints.

5. Temporal variation and peak-hour effects: Temporal patterns in congestion are not uniform across vehicle types. Passenger car congestion remains relatively stable throughout the day with only modest variation, while heavy vehicles exhibit more pronounced time-of-day differences. This indicates that peak-hour effects are present but are more evident for specific vehicle categories rather than consistently across all traffic.

h. Policy Implications

The findings have several important implications for transportation policy and planning:

Targeted interventions at identified hotspots: The observed spatial concentration of congestion and emissions suggests that a limited number of locations may account for a substantial share of the observed traffic and environmental impacts. Chronic congestion hotspots, where elevated congestion persists across multiple hours of the day, may therefore represent useful areas for further investigation.

Heavy vehicle management: The differential impact on heavy vehicles suggests potential benefits from truck-specific measures such as dedicated lanes, off-peak delivery incentives, or dynamic speed management. Such measures may be worth exploring as potential means of improving both freight efficiency and environmental performance, pending further targeted study.

Congestion pricing and demand management: The non-linear relationship between congestion and emissions implies that reducing peak demand, even modestly, can yield disproportionate environmental benefits. Time-of-day pricing or peak spreading strategies may represent areas warranting further investigation, given the observed non-linear relationship between congestion severity and emissions.

Real-time traffic management: The instability observed in the transitional flow regime suggests opportunities for active traffic management systems to prevent breakdown conditions. Early intervention when approaching capacity thresholds may potentially reduce the risk of more severe congestion propagating downstream, though this remains to be tested in operational conditions.

Integrated environmental assessment: The descriptive findings suggest that congestion reduction and emission reduction may be complementary objectives in most observed cases. The findings suggest that incorporating quantitative emission impact assessment alongside traditional mobility metrics may provide additional value in transportation planning.

9. Discussion

The current data pipeline operates by accepting user-provided URLs stored in an input file, from which traffic data is retrieved, parsed, and processed. This approach can be further enhanced by enabling dynamic source discovery, whereby URLs are automatically identified and collected based on geographic location, road segments, or other user-defined parameters. Additionally, the implementation of a fully automated ETL (Extract, Transform, Load) process would facilitate continuous data ingestion into a centralized data warehouse, ensuring improved data consistency, scalability, and timeliness.

To enhance usability and accessibility, a web-based interface could be developed to streamline the management of input sources and monitor data extraction workflows. Building upon the data warehouse layer, an interactive analytical dashboard could be introduced, providing visualization tools and drill-down capabilities. Such a dashboard would enable stakeholders to explore congestion metrics, compare spatial and temporal patterns, and derive actionable insights to support data-driven transportation planning and decision-making.

The system also offers substantial potential for future development through the integration of advanced data warehousing technologies. Additional data sources may be incorporated, and extended database schemas can be designed to support multidimensional analysis. In the long term, implementing a dedicated data warehouse architecture such as a star or snowflake schema would facilitate efficient storage of large-scale historical traffic data and optimize analytical query performance. This architectural enhancement would support high-volume data

processing, improve scalability, and enable advanced analytics, including long-term trend analysis and congestion pattern detection.

Several extensions to data analysis can further improve the proposed framework. First, traffic microsimulation models can be integrated. This would allow predictive congestion analysis and scenario testing. Such models could allow policymakers to explore potential effects of infrastructure changes before implementation. Second, real-time emission measurements can be incorporated. This would strengthen the environmental assessment component. Linking congestion data with live environmental monitoring would improve sustainability analysis. Finally, future research may expand the framework to additional regions and road categories.

10. Conclusion

This study set out to address two research questions concerning the extraction, structuring, and analysis of historical traffic sensor data from the Swedish Transport Administration (Trafikverket).

The results of the implementation indicate that the proposed data pipeline is capable of parsing historical traffic data from Trafikverket into a structured relational format. Operating on a system with a 2.6 GHz Intel Core i5 CPU and 24 GB of RAM, the extraction pipeline achieved an average ingestion rate of approximately 1.5 rows per second under standard network conditions. For experimental validation, the application was run for a short duration, during which more than 5,000 records were stored within a brief time frame. Repeated executions of the program resulted in zero duplicate insertions, suggesting that the deduplication mechanism functions reliably under the tested conditions. Furthermore, the adherence of the stored data to a well-defined schema, combined with the manual verification results, provides strong evidence that the automated extraction and structuring process is effective. Consequently, these findings address the first research question by illustrating a technically viable approach to historical traffic data extraction and structured storage.

Regarding the second research question, the empirical analysis suggests that traffic conditions on the studied corridors are generally characterized by persistent

moderate congestion, with a mean Speed Ratio of 0.770. Congestion and emission intensity show spatial clustering, with certain locations consistently exhibiting lower Speed Ratios and higher emission estimates. From a temporal perspective, passenger car congestion remains relatively stable throughout the day, while heavy vehicles display more pronounced variation. Lower Speed Ratios, particularly under severe congestion conditions ($SR < 0.6$), were generally associated with higher estimated per-vehicle CO₂ emissions based on the COPERT methodology. Taken together, these findings offer a descriptive basis for identifying high-impact congestion locations and illustrate the potential value of vehicle-type-specific traffic analysis in supporting transportation planning. This analysis addresses the second research question.

Overall, the proposed framework integrates data engineering, congestion modeling, and environmental assessment into a reproducible workflow. The findings are descriptive in nature and are based on a selected set of measurement points. Causal interpretation and generalization to the broader Swedish road network would require further research. Nevertheless, the study demonstrates how publicly available historical traffic sensor data can be transformed into structured analytical outputs that may support evidence-based transportation planning.

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Appendix A

a. Results from Analysis

Indicator Validation

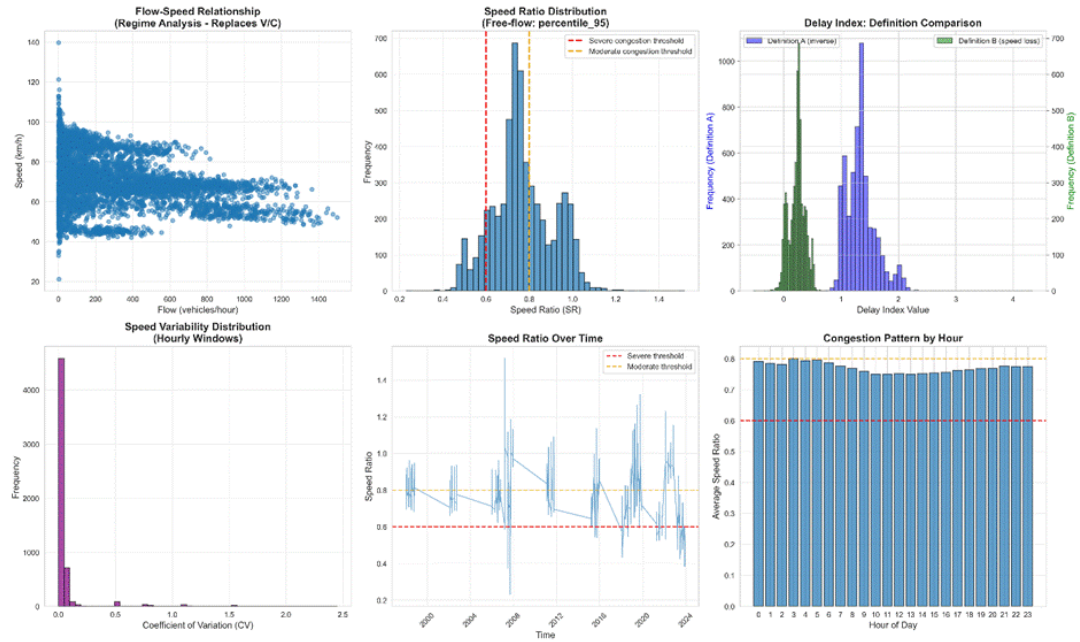


Figure 10: Traffic Congestion Indicator Validation Diagnostics

The six-panel diagnostic assessment validates the congestion indicators used in this study. The flow-speed scatter plot reveals three distinct traffic regimes (uncongested, transitional, and congested), with empirical capacity estimated at ~1,500 vehicles/hour. The Speed Ratio distribution shows 64.7% of observations in moderate-to-severe congestion ($SR < 0.8$), with a mean of 0.770. The Delay Index comparison confirms that Definition B ($DI = 1 - v_{obs}/v_{ff}$) produces a normalized speed-loss metric that quantifies the proportional reduction in speed relative to the percentile-based free-flow reference. This differs from traditional travel-time-ratio definitions of delay. While most values fall within the 0-1 range under typical traffic conditions, negative values occur when observed speeds exceed the estimated free-flow reference. These negative values were retained in the analysis to preserve the full distribution of observed traffic conditions and to maintain consistency with the speed-based formulation. Speed variability analysis shows 96.2% of observations exhibit stable flow ($CV < 0.3$). The temporal analysis reveals minimal peak-hour

effects, with hourly mean SR varying only from 0.74 to 0.79 across the 24-hour cycle.

Temporal Patterns

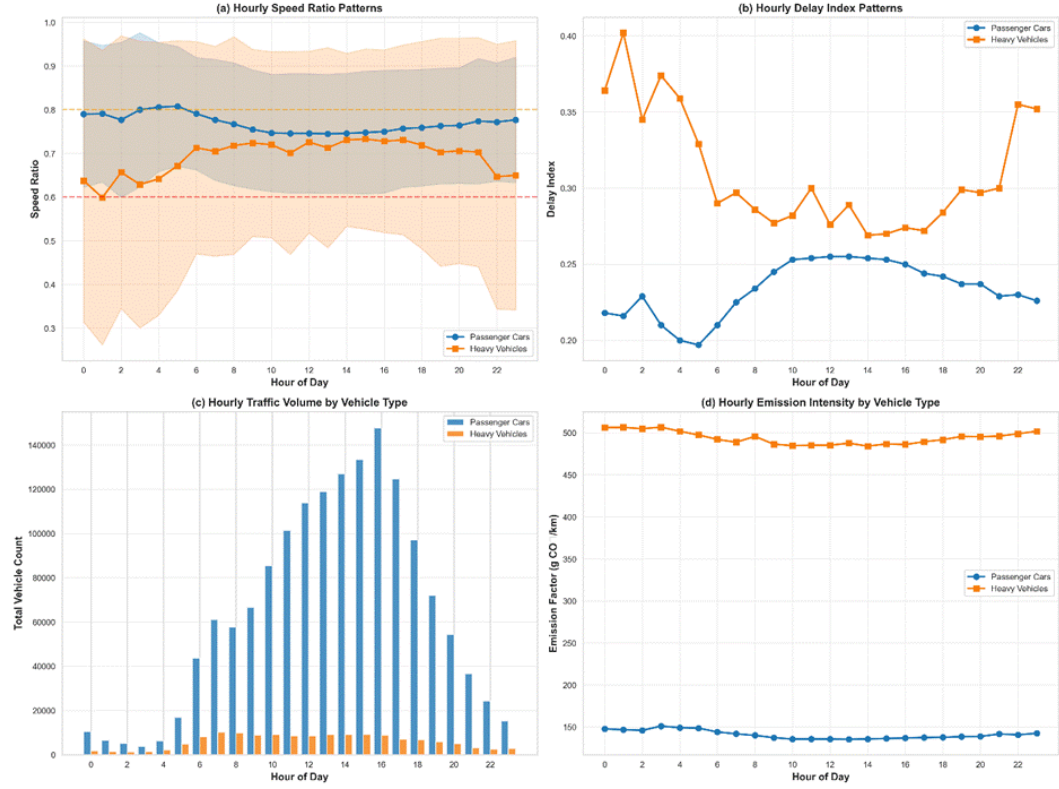


Figure 11: Hourly Congestion, Volume, and Emission Patterns

The four-panel temporal analysis reveals distinct patterns by vehicle type. Panel (a) shows passenger cars maintain stable SR around 0.78-0.80 throughout the day, while heavy vehicles exhibit pronounced early-morning degradation (SR = 0.62-0.65 at hours 1-5) and midday recovery (SR = 0.71-0.73). Panel (b) demonstrates this translates to a 38-40% travel time penalty for heavy vehicles during overnight hours. Panel (c) displays traffic volume patterns with passenger car peaks during afternoon hours (13:00-17:00) and more uniform for heavy vehicle distribution. Panel (d) shows emission factors remain relatively stable across hours: passenger cars at 145-150 g CO₂/km and heavy vehicles at 490-510 g CO₂/km (3.3-3.5× higher).

Key Finding: Unlike typical urban traffic patterns, the data from these rural/highway corridors suggest minimal passenger car peak-hour effects, with more pronounced variation observed for heavy vehicles during nighttime hours.

Spatiotemporal Heatmaps: Delay Index

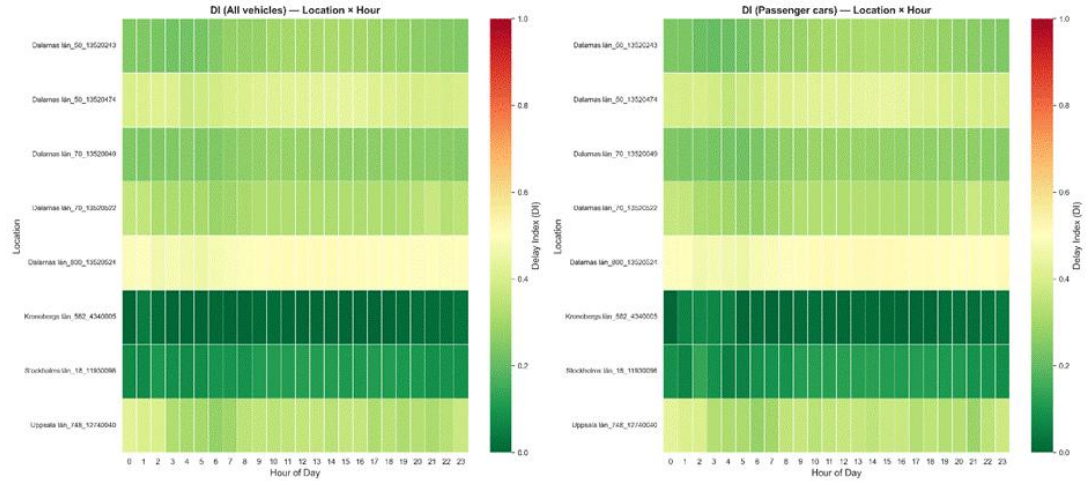


Figure 12: All Vehicles and Passenger Cars (Location × Hour)

The side-by-side heatmaps show DI values ranging from dark green ($DI \approx 0.0$) to dark red ($DI \approx 1.0$). Kronobergs län_582 and Stockholms län_18 display uniformly dark green coloring ($DI < 0.10$) across all hours, while Dalarnas län_800 shows uniform yellow ($DI \approx 0.40$ - 0.50), indicating chronic moderate congestion. The dominant pattern is horizontal uniformity within each location rather than vertical banding across hours, confirming that spatial factors outweigh temporal factors.

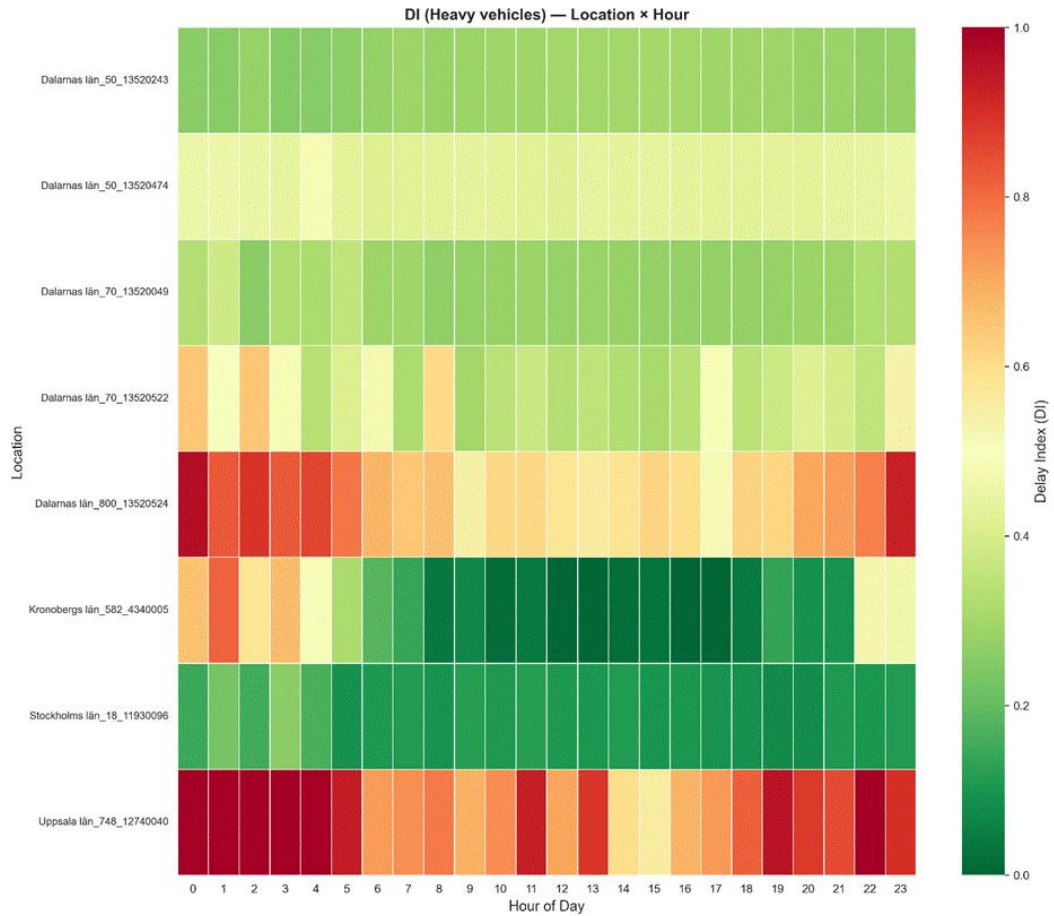


Figure 13: Heavy Vehicles (Location × Hour)

The heavy vehicle heatmap reveals substantially different patterns. Uppsala län_748 displays deep red/maroon coloring across all 24 hours ($DI \approx 0.80-1.00$), representing the worst conditions in the dataset. Dalarnas län_800 exhibits pronounced temporal banding with dark red during hours 0-6 and lighter orange to red during daytime. Several locations show counterintuitive nighttime degradation for heavy vehicles.

Key Finding: Heavy vehicles experience fundamentally different congestion patterns than passenger cars, with extensive temporal variability and severe nighttime episodes at specific locations

Delay Index Detailed Analysis

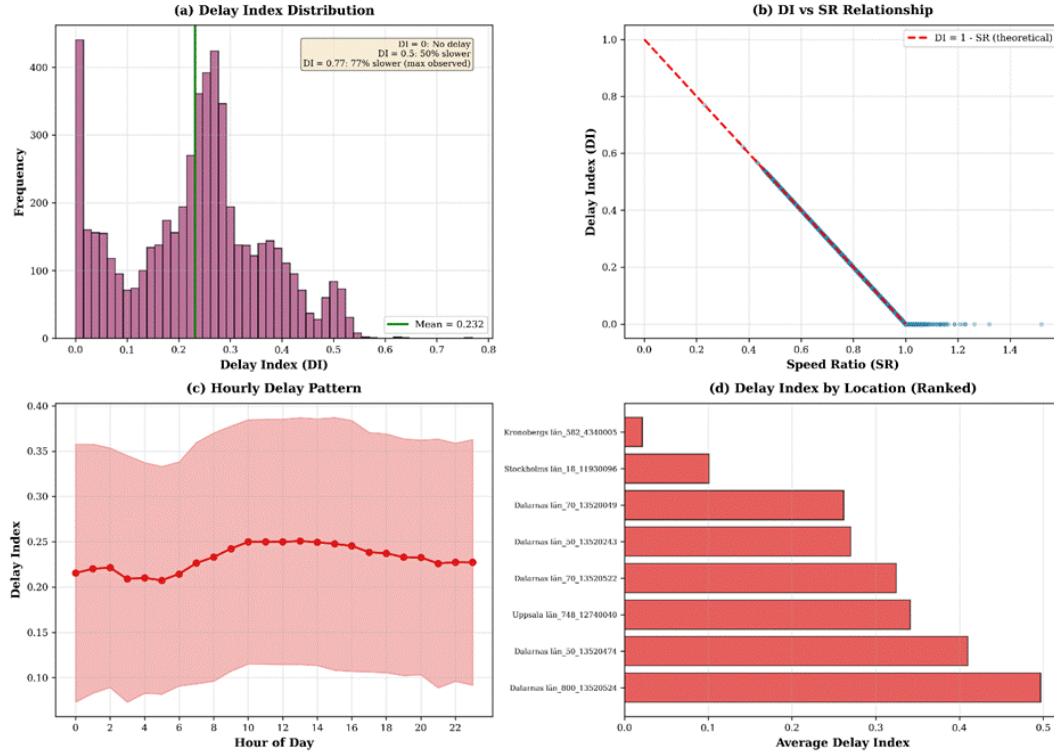


Figure 14: Distribution and Patterns (4-panel)

Panel (a) displays a bimodal DI distribution with mean = 0.232 and maximum observed DI = 0.77 (indicating speeds reduced to approximately 23% of the free-flow reference). Panel (b) validates perfect alignment with the theoretical relationship $DI = 1 - SR$. Panel (c) shows minimal hourly variation in mean DI (0.21-0.25 across 24 hours) but large constant spatial range (~0.30 DI points). Panel (d) ranks locations from Kronobergs_582 (DI = 0.05) to Dalarnas_800 (DI = 0.52), representing a 47-percentage-point performance gap.

Key Finding: Spatial variation appears to dominate temporal variation in this dataset, with a 10-fold performance difference between best and worst locations.

Vehicle Type Comparison

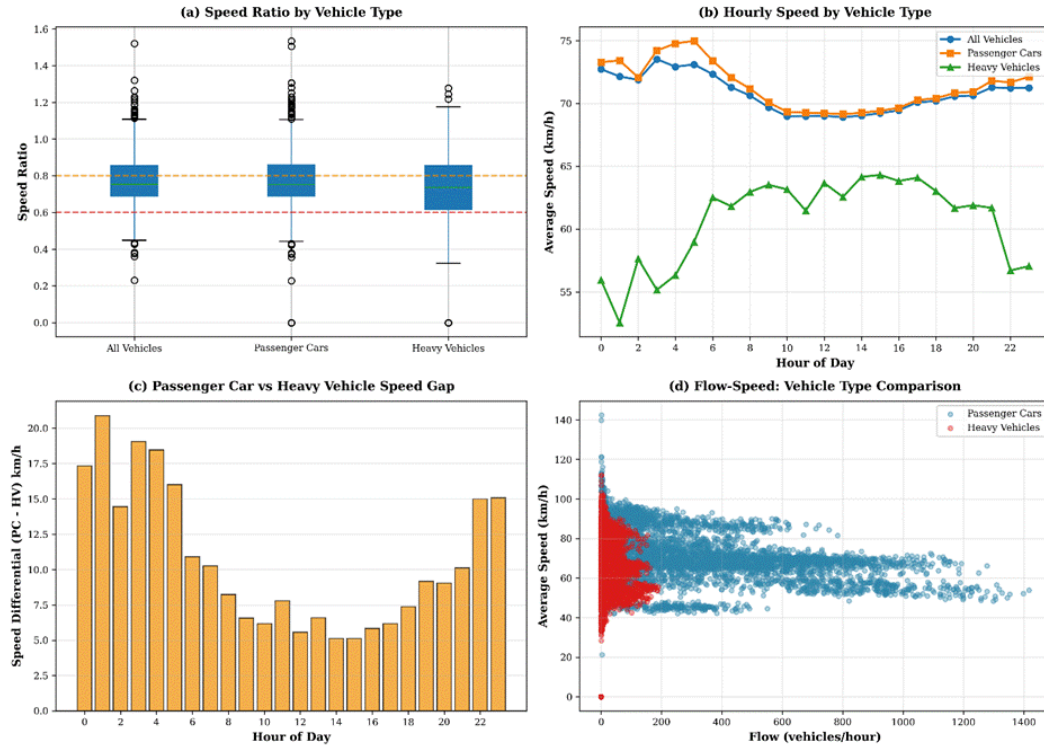


Figure 15: Speed Patterns and Flow Relationships (4-panel)

Panel (a) shows box plots with passenger car median SR ≈ 0.80 versus heavy vehicle median SR ≈ 0.65 , representing a systematic 0.12-0.15-point disadvantage. Panel (b) traces hourly speeds, revealing passenger cars maintain stable 69-75 km/h while heavy vehicles show pronounced variation (52-64 km/h) with a persistent 7-20 km/h gap. Panel (c) quantifies this gap by hour, showing a maximum disparity of 17-21 km/h during late night and a minimum of 5-7 km/h during midday. Panel (d) presents flow-speed scatter plots showing complete separation between passenger car and heavy vehicle point clouds, confirming distinct performance regimes.

Key Finding: Heavy vehicles may operate in a fundamentally different performance regime, with speed gaps varying by a factor of 3-4 across the day, with a minimum during midday business hours (5-7 km/h).

Hourly Emission Patterns by Vehicle Type

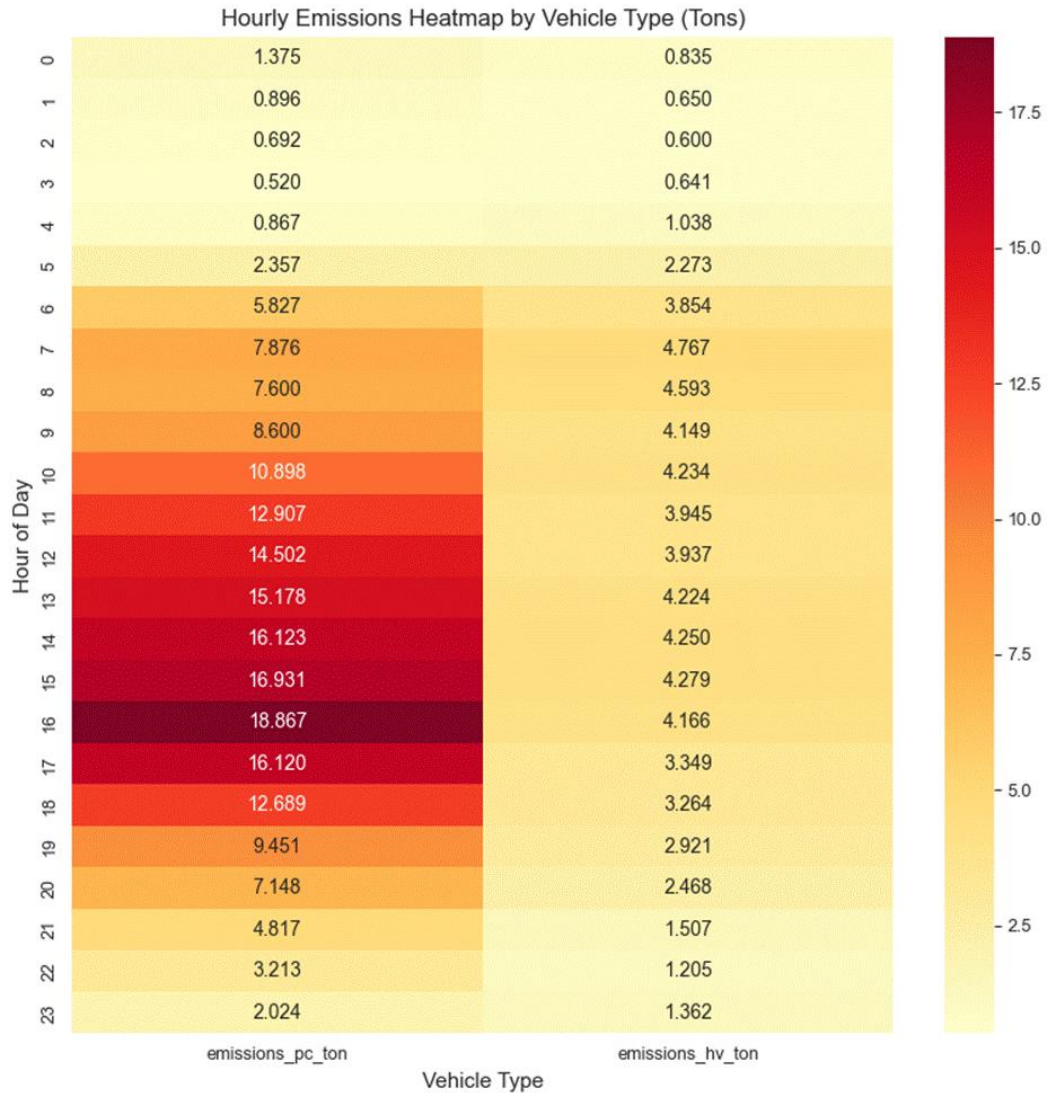


Figure 16: Hourly Emissions Heatmap by Vehicle Type (Tons)

The heatmap displays total CO₂ emissions (tons) aggregated across all locations for each hour of the day, separated by vehicle type.

Passenger Cars: Emissions range from 0.520 tons (hour 3, overnight minimum) to 18.867 tons (hour 16, afternoon peak). The pattern shows clear diurnal variation with three distinct phases: overnight low-emission period (hours 0-5: 0.52-2.36 tons), morning increase (hours 6-9: 5.83-8.60 tons), and sustained afternoon-evening peak (hours 10-18: 10.90-18.87 tons) with maximum at hour 16. Evening decline begins after hour 18, decreasing to 2.02 tons by hour 23.

Heavy Vehicles: Emissions range from 0.600 tons (hour 2) to 4.767 tons (hour 7). The temporal pattern differs substantially from passenger cars, showing more

uniform distribution across business hours. Morning hours (6-10: 3.85-4.59 tons) align with freight delivery schedules, with peak at hour 7 (4.77 tons). Daytime emissions remain elevated (hours 11-16: 3.95-4.28 tons) before gradual evening decline. Overnight emissions (hours 0-5: 0.60-2.27 tons) are proportionally higher than passenger cars, reflecting continuous freight operations.

Key Finding: Passenger car emissions exhibit pronounced peak-hour concentration (afternoon peak 36× higher than overnight minimum), while heavy vehicle emissions show more sustained patterns throughout business hours (peak only 8× higher than minimum), reflecting fundamental differences in travel purposes—commuting versus commercial freight operations.

b. Results from the MCP server

A natural language query was submitted to the MCP (Model Context Protocol) server, which returned the available data options from the traffic database through the Traffic MCP server (Figure 17).

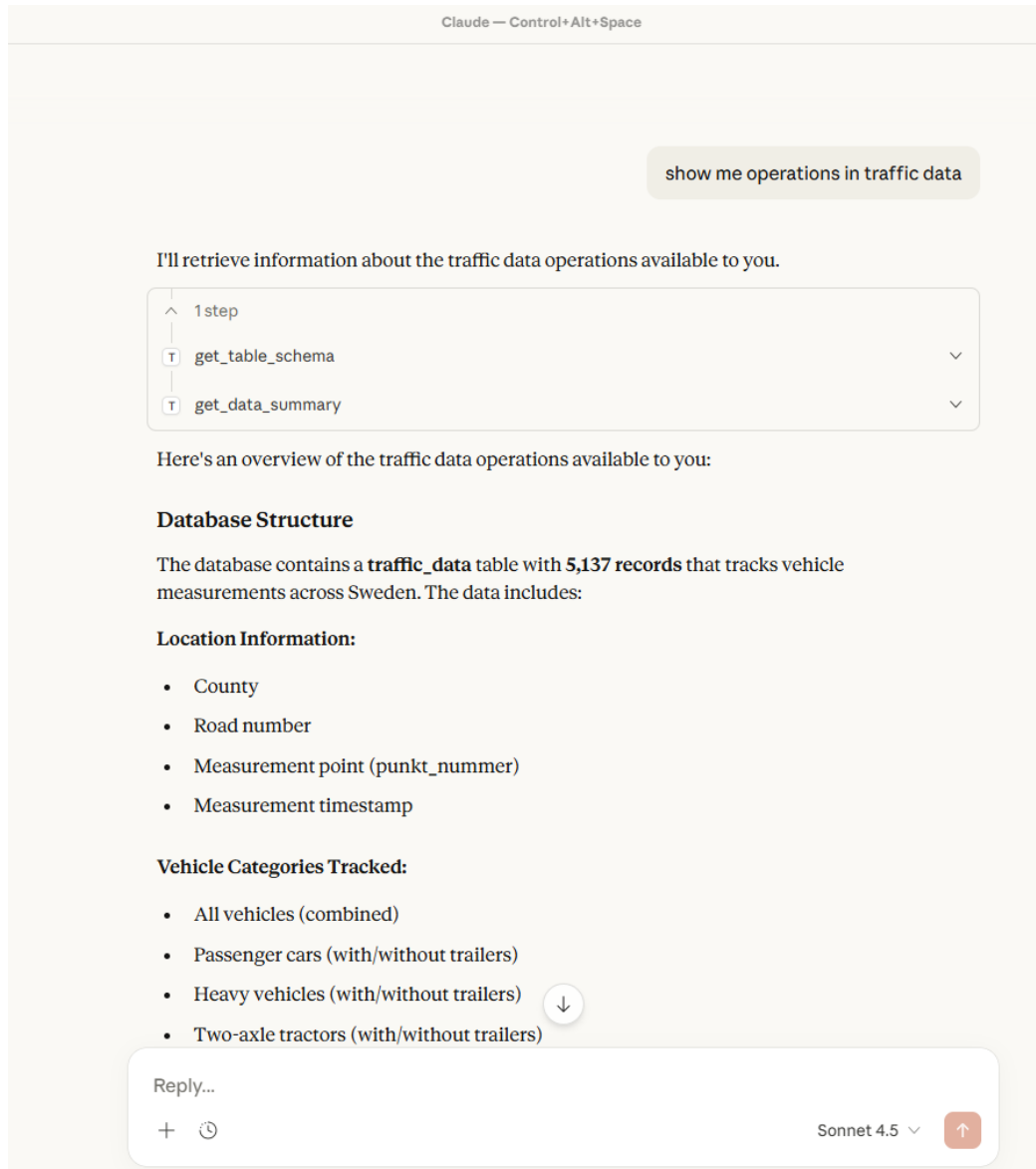


Figure 17: Operation listing in Claude Desktop

Since vehicle speed-related data are stored in the database, the integration of the AI client (Claude Desktop) with the Traffic MCP server was used to generate statistical analyses of vehicle speeds across different vehicle categories (Figure 18). The AI client also returned a summary of the observations based on the retrieved data (Figure 19).

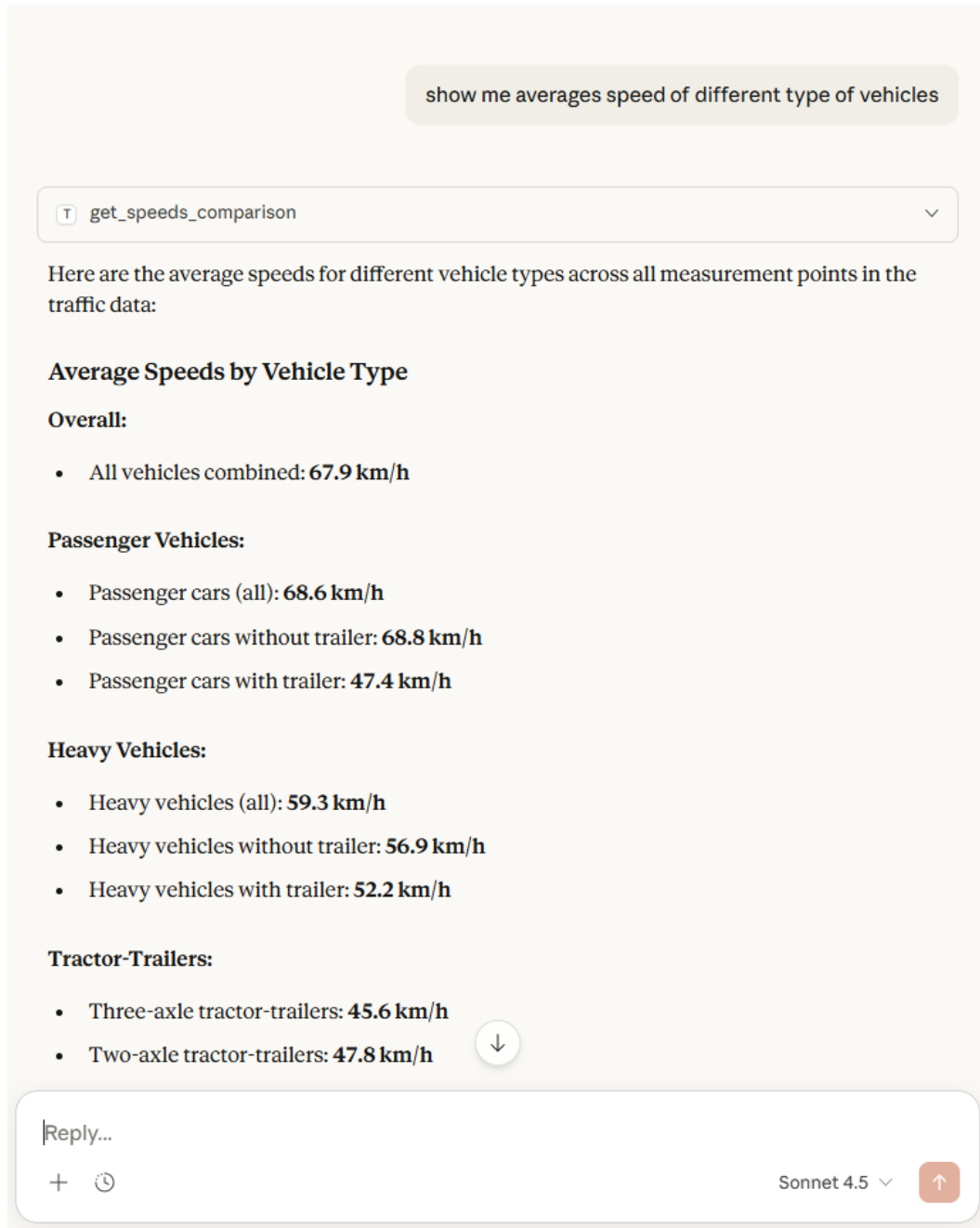


Figure 18: Claude output for chat to fetch average speeds

Key Observations

The data shows clear speed patterns:

- Passenger cars without trailers are the fastest at nearly 69 km/h
- Adding a trailer significantly reduces speed (21 km/h slower for passenger vehicles)
- Heavy vehicles travel about 9 km/h slower than passenger cars on average
- Large tractor-trailers are the slowest, averaging around 46-48 km/h

Would you like to see this data visualized in a chart, or would you like to filter by specific roads, counties, or time periods?



Reply...



Sonnet 4.5



Figure 19: Claude output for chat to fetch average speeds

Appendix B

Trafikverket Data Extractor

A Python-based web scraper that extracts traffic measurement data from the Trafikverket website and stores it in a PostgreSQL database.

<https://github.com/supal/trafficdata>

Features

- Automated web scraping - Automatically navigates the Trafikverket website
- Multi-URL support - Process multiple URLs from a configuration file
- PostgreSQL integration - Stores traffic data in a relational database with automatic schema creation
- Headless mode - Fast, background execution without UI
- Intelligent waits - Executes as soon as elements load, not fixed delays
- Duplicate detection - Prevents duplicate records based on composite keys
- Python 3.9+ compatible - Works with Python 3.9.6 and newer versions

Prerequisites

- Python 3.9.6 or higher
- Google Chrome (must be installed on your system)
- PostgreSQL 12+ (local or remote database server)
- pip (Python package manager)
- git (for version control, optional)

Installation for Fresh Environment

Step 1: Clone or Download the Project

```
# If using git
git clone https://github.com/supal/trafficdata.git
cd trafficdata

# Or download and extract the ZIP file manually
```

Step 2: Set Up PostgreSQL Database

Create the database and user (use default credentials for local development):

```
# Login to PostgreSQL
psql -U postgres

# Create database and user
CREATE DATABASE traffic_data;
CREATE USER trafficdata WITH PASSWORD 'trafficdata';
ALTER ROLE trafficdata SET client_encoding TO 'utf8';
ALTER ROLE trafficdata SET default_transaction_isolation TO 'read
committed';
ALTER ROLE trafficdata SET timezone TO 'UTC';
GRANT ALL PRIVILEGES ON DATABASE traffic_data TO trafficdata;

# Exit psql
\q
```

Or edit scraper.py to customize database credentials:

```
self.db_config = {
    'host': 'your-host',
    'user': 'your-username',
    'password': 'your-password',
    'database': 'traffic_data',
    'port': 5432
}
```

Step 3: Set Up Virtual Environment and Dependencies

On macOS/Linux:

```
# Make scripts executable
chmod +x run.sh setup.sh

# Run setup script (creates venv and installs dependencies)
./setup.sh
```

On Windows:

```
# Option 1: Use provided batch script
setup.bat

# Option 2: Manual setup
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
```

Step 4: Verify Installation

```
# On macOS/Linux
./run.sh scraper.py --help

# On Windows
run.cmd scraper.py --help
```

You should see the help menu with available options.

Quick Start

1. Ensure PostgreSQL is Running

Make sure your PostgreSQL server is running, and the traffic_data database is created.

```
# On macOS with Homebrew
brew services start postgresql

# On Linux with systemd
sudo systemctl start postgresql

# On Windows
# PostgreSQL service should start automatically, or start it from
Services
```

2. Add URLs to Process

Edit input_url.txt and add one URL per line:

```
https://vtf.trafikverket.se/tmg101/AGS/tmg104bestaellinfouttag.aspx?punk
tnrlista=13520522,13520522&laenkrollista=2,3
https://vtf.trafikverket.se/tmg101/AGS/tmg104bestaellinfouttag.aspx?punk
tnrlista=13520474,13520475&laenkrollista=1,1
```

3: Run the Scraper

Recommended (fastest):

```
./run.sh scraper.py --headless # macOS/Linux
run.cmd scraper.py --headless # Windows
```

With browser window visible (for debugging):

```
./run.sh scraper.py # macOS/Linux
run.cmd scraper.py # Windows
```

4: Check Output

Data will be automatically stored in the PostgreSQL traffic_data database. Query the data using:

```
# Connect to the database
psql -U postgres -d traffic_data

# Query all records
SELECT * FROM public.traffic_data;

# Query by date range
SELECT * FROM public.traffic_data
WHERE measurement_time >= '2026-01-01' AND measurement_time < '2026-02-
01';

# Query by road number
SELECT * FROM public.traffic_data WHERE road_number = '25';
```

Command-Line Options

Usage: scraper.py [-h] [-u URL] [--headless]

Options:

-h, --help	Show help message and exit
-u, --url URL	Single URL to process (overrides input file)
--headless	Run browser in headless mode (faster, no UI)

Usage Examples

Process all URLs from input_url.txt in headless mode (fastest):

```
./run.sh scraper.py --headless
```

Process a single URL:

```
./run.sh scraper.py -u "https://vtf.trafikverket.se/..." --headless
```

Run with browser window visible (for debugging):

```
./run.sh scraper.py
```

Output Format

Data is stored directly in the PostgreSQL traffic_data database with the following schema:

Table: `public.traffic_data`

Column	Type	Description
id	SERIAL PRIMARY KEY	Unique record identifier
measurement_time	TIMESTAMP	Date and time of measurement
county	VARCHAR(100)	County/Region (Län)
road_number	VARCHAR(10)	Road number (Vägnr)
punkt_nummer	VARCHAR(20)	Measurement point ID

Column	Type	Description
all_vehicles_count	INTEGER	Total vehicle count
all_vehicles_avg_speed	DECIMAL(5, 2)	Average speed for all vehicles
passenger_car_count	INTEGER	Passenger car count
passenger_car_avg_speed	DECIMAL(5, 2)	Average passenger car speed
heavy_vehicles_count	INTEGER	Heavy vehicle count
heavy_vehicles_avg_speed	DECIMAL(5, 2)	Average heavy vehicle speed
*_trailer_count / *_no_trailer_count	INTEGER	Breakdown by trailer type
*_trailer_avg_speed / *_no_trailer_avg_speed	DECIMAL(5, 2)	Speed data by trailer type
created_at	TIMESTAMP	Record insertion timestamp

Key Features:

- Automatic schema creation on first run
- Composite key duplicate detection (measurement_time, county, road_number, punkt_nummer)
- Indexed for fast queries on measurement_time, punkt_nummer, and road/county combinations

Project Structure

```
trafficdata/
├── scraper.py          # Main scraper application
```

```

├─ cli.py                # Command-line interface (alternative entry
point)
├─ compatibility.py      # Python version validation
├─ input_url.txt         # URLs to process (one per line)
├─ run.sh               # Helper script for macOS/Linux
├─ run.cmd              # Helper script for Windows
├─ setup.sh             # Setup script for macOS/Linux
├─ setup.bat            # Setup script for Windows
├─ requirements.txt      # Python dependencies
├─ requirements-py39.txt # Python 3.9 specific dependencies
├─ .python-version      # Python version hint
└─ README.md           # This file

```

How It Works

For each URL:

- **Navigate** - Opens the Trafikverket webpage
- **Detect** - Finds all measurement occasions available
- **Loop** - For each measurement occasion:
 - Select the occasion from the dropdown
 - Check all data checkboxes
 - Select table format
 - Click the start button
 - Wait for the pop-up window
 - Extract data from the table
- **Insert** - Stores each record in PostgreSQL with duplicate detection
- **Commit** - Data has persisted in the database

Dependencies

Python Packages

- selenium (4.38.0) - Web browser automation
- pandas (2.3.3) - Data manipulation
- psycopg2 (2.9.0+) - PostgreSQL database adapter
- webdriver-manager (4.0.2) - Automatic Chrome driver management

All dependencies are automatically installed during setup.